The geometry of AI validation: Exact certification limits for iid best-of-N search

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Abstract

Abstract. AI systems increasingly generate alternatives, inspect evidence, and deploy a selected output. Validation is therefore target-relative: evidence certifies deployment only in directions resolved by the interventions that produced it. We represent validation and deployment rules as kernels over a reliability surface. Their span geometry separates two resources: replication reduces sampling noise, whereas new intervention directions reduce structural blindness. We make this principle exact in the canonical iid best-of-N problem. For iid candidates, scalar ranking, randomized ties, maximum selection, bounded binary truth, and an unchanged rank–truth relation, knowing best-of-n reliability through $n = m$ leaves exact width

$$B_{m,N} = 1 + 2\sum_{r=1}^{m}(-1)^r \cos^{2N}\left\{\frac{r\pi}{2(m+1)}\right\},$$

Explicit bounded worlds attain the whole interval, and this prefix is information-maximal among reliability-mean audits confined to $n \leq m$. The governing scale is m^2/N: at $m \sim \sqrt{N}$, ambiguity is about 0.83; in the iterated fixed-phase then small-ε limit, width ε requires $m \asymp \sqrt{N \log(1/\varepsilon)}$. Monotonicity gives an exact uniform-approximation frontier; a Lipschitz bound gives an exact capped-tail dual and order-sharp L/m^2 ambiguity. These results yield a two-gate audit rule: establish structural coverage, then add independent tasks for precision. Retrospective CodeRM programs construct compatible best-of-100 values spanning $[0.186, 0.992]$ and $[0.305, 0.996]$ as feasible witnesses, and a score-tail rule frozen after 82 discovery tasks reduced 500-label held-out 95th-percentile error from 0.180 to 0.049 and from 0.076 to 0.035. Beyond iid search, the geometry applies only to known or independently estimated kernels. The empirical analyses are illustrative, not prospective interventions.

Keywords: AI evaluation; inverse problems; post-selection inference; optimal experimental design; inference-time search.

Artificial-intelligence systems increasingly do more than return a fixed output. They generate alternatives, compare them, and act on a selected answer, program, prompt, agent, molecule, experiment, or scientific hypothesis [1–3]. The act of selection changes the population whose truth matters. Validation is therefore target-relative: reliability belongs to the inference produced by a deployment rule, not to a model as an intrinsic scalar.

This paper begins from one geometric claim. Each validation intervention defines a kernel describing where it looks; deployment defines another kernel describing where it will place weight. The validated kernels span the directions resolved by evidence. If a deployment target lies outside that span, two reliability worlds can agree on every audit and still disagree after selection. That residual is *structural blindness*. More observations inside the same interventions reduce noise but do not change the span; a new intervention can. If the target is chosen after evidence is inspected, validity must additionally protect the selectable family. Structural identification, post-selection inference, and evaluation design are thus consequences of the same target-relative geometry.

We make that geometry exact in a canonical deployment problem. Suppose an evaluator knows, without sampling error, iid best-of-n reliability for every $n = 1, \ldots, m$, under a stable rank–truth law, scalar ranking, randomized ties, and maximum selection, but deployment searches $N > m$ candidates. How far apart can two systems be at N while agreeing at every audited width? The answer is Eq. 14; explicit worlds attain the whole interval. At $m \sim \sqrt{N}$, its limiting width remains approximately 0.83, and small ambiguity ε requires $m^2/N \sim \log(1/\varepsilon)$.

The argument follows one idea through four consequences. A target outside the validated span creates exact structural ambiguity. A target selected after evaluation requires family-level rather than unrelated pointwise validity. Audit design should therefore add feasible directions that expand the span. In iid best-of-N search, the geometry becomes a closed finite frontier, an information-maximality law for low-width reliability means, and a finite audit-depth rule. Shape constraints test whether the phenomenon survives monotonicity and smoothness; the finite-task planner and two objective-domain studies test how structural breadth interacts with task precision and deployment-matched labels. These are not separate warnings: they ask whether one selection direction is resolved by the interventions used to validate it.

Optimal recovery, truncated Hausdorff moments, post-selection inference, and polynomial approximation supply established ingredients [4–9]. The mathematical contribution is the validation-specific package obtained after those ingredients are specialized: a full attainable bounded-binary identified interval with explicit common-audit worlds; maximality of the complete prefix within the reliability-mean query class and its finite theta phase law; and exact monotone and capped-Lipschitz residual frontiers linked to a finite audit plan. We isolate the canonical L^1 calculation as classical input C0. SI Appendix, Section S11 compares each submitted object with the adjacent moment, expected-order-statistic, and best-of-N literatures; no broader theorem about moment spaces or optimal recovery is asserted.

The scope is equally explicit. The closed form $B_{m,N}$ applies only to iid candidates, scalar ranking, randomized ties, maximum selection, bounded truth, and a stable rank–truth law. The span theorem covers a different selection mechanism only when its audit and deployment kernels act on a common stable state space and are known or independently estimated. Unmodeled adaptive generation receives no certificate from either result. Here m measures coverage of fixed search widths, not labels, tasks, generations, or monetary cost.

Validation is a target-relative inverse problem

The geometry has three objects: f describes where the system is reliable; each v_j describes where a validation procedure looks; and k describes where deployment will place weight. If k emphasizes a direction that no combination of the v_j observes, exact repetition inside the existing tests cannot recover the missing target. The best-of-N result will make this general principle fully explicit for one important order statistic.

Let $(\mathcal{X}, \mu)$ describe latent deployment conditions—for example, score ranks, prompts, task subpopulations, or workflow states. Let

$$f(x) = \mathbb{P}(T = 1 \mid X = x), \qquad 0 \le f \le 1, \tag{1}$$

be the unknown reliability surface. A normalized validation intervention $v_j \in L^1(\mu)$ observes

$$y_j = \langle v_j, f\rangle := \int v_j(x) f(x)\, d\mu(x), \qquad \int v_j\, d\mu = 1. \tag{2}$$

A deployment choice w induces another normalized kernel k_w and target

$$R_w(f) = \langle k_w, f\rangle. \tag{3}$$

Kernels may be probability densities, signed contrasts, or known importance weights; normalization is used only to give the central evidence below the simple value 1/2.

Write $S_V = \operatorname{span}\{v_1, \dots, v_p\} \subset L^1(\mu)$ and define the blind residual

$$b(k; V) = \operatorname{dist}_{L^1}(k, S_V) = \inf_{s \in S_V} \|k - s\|_1. \tag{4}$$

This distance has an exact inferential meaning.

Theorem 1 (Span theorem and sharp ambiguity). *For* $\mathcal{F} = \{f \in L^\infty(\mu) : 0 \le f \le 1\}$ *and validation operator* $Kf = (\langle v_j, f\rangle)_{j=1}^p$*:*

(i) $R_k(f)$ *is identified by* Kf *for every* $f \in \mathcal{F}$ *if and only if* $k \in S_V$.

(ii) The largest separation between two validation-equivalent worlds is

$$\sup_{\substack{f,g \in \mathcal{F}\\ Kf = Kg}} |R_k(f) - R_k(g)| = b(k; V). \tag{5}$$

(iii) If all kernels integrate to one, the evidence $Kf = (1/2, \dots, 1/2)$ *has an exact identified interval of width* $b(k; V)$*, centered at* $1/2$*.*

The theorem formalizes the mental model. Validation-equivalent worlds can differ only in directions orthogonal to every v_j; pairing the deployment kernel with the largest bounded such difference gives exactly its distance from the validated span. The quotient-duality and extremizer argument appears in SI Appendix, Section S1. The equality is important: Eq. 5 is the attainable ambiguity, not only an upper bound.

Generality at theorem level, not formula level. A randomized shortlist, top-k draw, rejection sampler, beam-search trace, or dependent candidate process induces a selected-state law different from Nu^{N-1}. If that law is known on a common stable state space, Theorem 1 still gives its exact structural ambiguity as a distance from the audited span, without requiring iid candidates. It does not give that mechanism the best-of-N value $B_{m,N}$. For an adaptive generator, the state must contain the relevant history and the policy-induced kernel must be known or independently estimated; if either is missing, neither the general theorem nor the iid frontier supplies a certificate. When those kernels are estimated with L^1 error bounds, SI Appendix, Section S7.2 gives an explicit perturbation penalty weighted by the reconstruction coefficients. Thus the broad result is a known- or estimated-kernel geometry, whereas the closed-form phase law is the exact iid specialization.

For a selectable family $\mathcal{K} = \{k_w : w \in \mathcal{W}\}$, define

$$B(\mathcal{K}; V) = \sup_{w \in \mathcal{W}} b(k_w; V). \tag{6}$$

This is the structural half of validation capacity. If $B = 0$, every target in the family is reconstructible from the validated interventions. If B is near one for probability kernels, some selectable target is compatible with nearly opposite reliability under the same exact evidence. Importantly, increasing the number of observations inside any existing v_j does not change S_V and hence cannot change B.

Selection after evidence requires family-level validity

Pointwise identification and pointwise confidence are not enough when deployment is chosen after evaluation. For a finite target family, let $C_w(Y)$ be an interval for $R_w(f)$ and let $\widehat{w}(Y)$ be any measurable selector.

Proposition 1 (Arbitrary selection equals simultaneous protection). *A confidence system is valid at level* $1 - \alpha$ *after* every *measurable target-selection rule if and only if it covers all targets simultaneously:*

$$\inf_{f \in \mathcal{F}} \mathbb{P}_f\{R_w(f) \in C_w(Y) \text{ for every } w \in \mathcal{W}\} \ge 1 - \alpha. \tag{7}$$

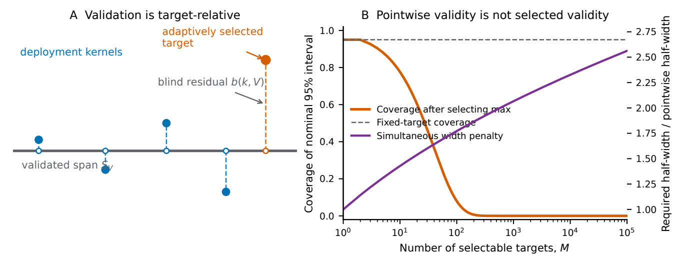

Figure 1: Target-relative and post-selection geometry. (A) Validation kernels span S_V. A deployment kernel decomposes into a represented component and a blind residual whose exact L^1 norm is the structural ambiguity. Post-evaluation selection can choose a kernel with a large residual. The drawing is schematic; the theorem uses L^1/L^∞ quotient duality. (B) For M independent Gaussian target estimates, every fixed-target interval has 95% coverage, but the same interval around the largest estimate rapidly loses coverage. Simultaneous protection widens only as $\sqrt{\log M}$ in this example, yet no amount of replication removes a nonzero blind residual.

The necessity follows by selecting the first target whose interval misses, if one exists. This is the same logical reduction used in universal post-selection inference [6]; it is included to identify the correct unit of an AI validation claim, not as a new abstract post-selection theorem. A fixed, stable, or independently chosen selector can require less protection [7, 10]. An unrestricted search system cannot be certified by a collection of unrelated pointwise statements.

Figure 1B gives the elementary Gaussian illustration: ordinary 95% intervals have correct fixed-target coverage, but coverage collapses after deploying the target with the largest estimate. This stochastic selection cost requires family-level protection. The blind residual is different: it survives even when sampling noise is exactly zero. Under Gaussian audit noise of order $n^{-1/2}$, SI Appendix, Section S3 shows that any linear reconstruction $s_w = \sum_j a_{wj} v_j$ admits a bias-centered estimator satisfying, simultaneously for every w and every $f \in \mathcal{F}$,

$$|\widehat{R}_w - R_w(f)| \le \frac{1}{2}\|k_w - s_w\|_1 + \frac{q_{1-\alpha}(A, \Sigma)}{\sqrt{n}}. \tag{8}$$

The first term is structural and vanishes only when the deployment direction is represented. The second covers the selectable family and shrinks with replication. Thus replication changes sampling uncertainty, whereas a new intervention direction is needed to change structural blindness.

Expanding the validated span. The span theorem suggests choosing feasible new interventions—such as direct labels of selected winners or perturbed workflows—to minimize the largest blind residual over the deployment family. SI Appendix, Section S3.3 states the formal constrained-width objective. Redundant evaluations may improve precision while adding no new direction; one deployment-matched intervention can remove a residual that thousands of exchangeable items cannot.

Making the geometry exact: the finite best-of-N frontier

We now make the target-relative geometry fully explicit for the canonical score-and-select workflow. Let (U_i, T_i) be iid candidates, where the randomized score percentile satisfies $U_i \sim \mathrm{Unif}(0,1)$ and $T_i \in \{0,1\}$. Define

$$f(u) = \mathbb{P}(T = 1 \mid U = u). \tag{9}$$

For a score S with atoms, the precise randomized percentile is

$$U = F_S(S^-) + Z\{F_S(S) - F_S(S^-)\}, \qquad Z \sim \mathrm{Unif}(0,1) \tag{10}$$

with Z independent. Then U is uniform, while independence of the tie jitter makes f constant almost everywhere on the percentile interval occupied by each score atom. The unrestricted frontier remains exact over the stated model class because continuous-score candidate laws attain it; conditioning on a fixed known atomic score law adds interval-constancy restrictions and can only narrow the frontier. Best-of-n selects the largest percentile. Its reliability is

$$\theta_n = n \int_0^1 u^{n-1} f(u)\, du = \langle k_n, f\rangle, \qquad k_n(u) = n u^{n-1}. \tag{11}$$

At $n = 1$, k_1 weights every percentile equally. As n grows, k_n moves its mass into an increasingly thin upper tail: search has changed the population whose truth is being averaged. Exact audits at every $n = 1, \ldots, m$ span the polynomials $\mathcal{P}_{m-1}$. This is a bounded Hausdorff moment problem [8, 11]; the deployment target k_N is a more tail-concentrated order-statistic kernel [12, 13].

For a feasible audit vector $y = (y_1, \ldots, y_m)$, define the deployment identified set and its worst-case width by

$$\mathcal{I}_{m,N}(y) = \{\langle k_N, f\rangle : 0 \le f \le 1,\ \langle k_n, f\rangle = y_n\ (n \le m)\}, \tag{12}$$

$$W_{m,N} = \sup_{y:\, \mathcal{I}_{m,N}(y) \neq \varnothing} \operatorname{diam} \mathcal{I}_{m,N}(y). \tag{13}$$

Let U_m denote the degree-m Chebyshev polynomial of the second kind and define $\sigma_m(u) = \operatorname{sgn} U_m(2u - 1)$, with arbitrary values at its m roots.

Classical input C0 (Canonical L^1 distance (classical))**.** *For integers $N > m \ge 1$, canonical L^1 interpolation at the shifted roots of U_m gives*

$$\begin{aligned} B_{m,N} &:= \operatorname{dist}_{L^1}(k_N, \mathcal{P}_{m-1}) \\ &= 1 + 2\sum_{r=1}^{m} (-1)^r \cos^{2N}\left\{\frac{r\pi}{2(m+1)}\right\}. \end{aligned} \tag{14}$$

This approximation statement, including its nodes, error sign, and distance, is classical [14–18].

Theorem 2 (Finite-prefix certification theorem)**.** *For integers $N > m \ge 1$, the worst-case bounded-reliability identified width in Eq. 13 is exactly*

$$W_{m,N} = B_{m,N}. \tag{15}$$

Define two complete joint candidate laws $P_\pm$ by

$$\begin{aligned} U_\pm &\sim \mathrm{Unif}(0,1), \\ T_\pm \mid U_\pm = u &\sim \mathrm{Bernoulli}\{f_\pm(u)\}, \\ f_\pm(u) &= \frac{1 \pm \sigma_m(u)}{2}, \end{aligned} \tag{16}$$

and take candidate sequences iid under each law. These worlds satisfy $\theta_n^+ = \theta_n^- = 1/2$ for every $n \le m$, but

$$\theta_N^\pm = \frac{1 \pm B_{m,N}}{2}. \tag{17}$$

At the common evidence $(\theta_1, \ldots, \theta_m) = (1/2, \ldots, 1/2)$, the identified set is exactly

$$\left[\frac{1 - B_{m,N}}{2}, \frac{1 + B_{m,N}}{2}\right]; \tag{18}$$

every interior value is attained by the iid candidate law with $U \sim \mathrm{Unif}(0,1)$ and $T \mid U = u \sim \mathrm{Bernoulli}\{\lambda f_+(u) + (1-\lambda) f_-(u)\}$ for some $\lambda \in [0,1]$. Consequently Eq. 14 is the full worst-case identified width, not only a lower bound.

Classical input C0 computes a distance. Theorem 2 turns it into an attained certification statement: admissible worlds $(1 \pm \sigma_m)/2$ share one feasible audit vector, attain both deployment endpoints, and fill the interval by mixtures. The proof is in SI Appendix, Section S4; the mathematical boundary is documented separately in Section S11.

Proposition 2 (Information maximality of the complete low-width prefix)**.** *Consider any finite family of reliability-mean audits whose kernels are randomized mixtures of widths at most m,*

$$a_\ell = \sum_{n=1}^{m} q_{\ell n} k_n, \qquad q_{\ell n} \ge 0, \quad \sum_{n=1}^{m} q_{\ell n} = 1, \tag{19}$$

and let $S_A = \mathrm{span}\{a_\ell\}$. Its worst-case ambiguity at width $N > m$ is

$$\mathrm{dist}_{L^1}(k_N, S_A) \ge B_{m,N}. \tag{20}$$

Equality is attained by the complete prefix $k_1, \ldots, k_m$. The same lower bound holds for a sequential schedule that chooses each such mean audit after earlier audit means are observed.

Every kernel in Eq. 19 lies in $\mathcal{P}_{m-1}$, whereas the complete prefix spans that entire space. Sequential choice cannot distinguish the endpoint worlds because both return mean 1/2 for every admissible query. Thus the theorem is not merely about one convenient sweep of widths: among all exact population-mean schedules whose revealed kernels remain mixtures of fixed widths $n \le m$, the complete prefix is already the structurally strongest possible design. The claim excludes score-dependent stopping within a candidate run, finite-sample outcome-adaptive coefficient selection, direct width-N labels, and candidate-level rank–truth measurements; those can reveal a new direction or require uniform finite-sample control. The proof is in SI Appendix, Section S4.6.

The exact frontier has a nontrivial phase limit.

Corollary 1 (Search–validation phase law)**.** *If $m, N \to \infty$ with $m^2/N \to \tau \in (0, \infty)$, then*

$$B_{m,N} \longrightarrow B(\tau) := \vartheta_4\Big(0, e^{-\pi^2/(4\tau)}\Big). \tag{21}$$

Define $\tau_\varepsilon = \inf\{\tau > 0 : B(\tau) \le \varepsilon\}$. After first taking the joint limit at fixed τ and then letting $\varepsilon \downarrow 0$,

$$\tau_\varepsilon = \log\frac{1}{\varepsilon} + \frac{1}{2}\log\log\frac{1}{\varepsilon} + \log\frac{4}{\sqrt{\pi}} + o(1). \tag{22}$$

No uniform finite-(m, N) statement is asserted for an arbitrary sequence $\varepsilon = \varepsilon_N$.

The phase is sharper than the statement that m must be of order $\sqrt{N}$. At $m^2/N = 1$, the exact limiting width is approximately 0.83: auditing through $m \approx \sqrt{N}$ still permits reliability near opposite ends of the unit interval. Intuitively, m polynomial moments resolve the endpoint only on a scale of order m^{-2}, while best-of-N concentrates on a scale of order N^{-1}. Their ratio is m^2/N. The transformed theta series, both asymptotic regimes, and the derivation of Eq. 22 are in SI Appendix, Section S5 [19].

Robustness: the degree-squared law survives shape constraints

The exact frontier above is distribution-free. Its principal scientific stress test is whether blindness survives after ruling out oscillatory reliability. The Chebyshev worlds in Eq. 16 jump between zero and one; let $\mathcal{F}_\uparrow$ instead be the nondecreasing $[0,1]$-valued functions and, for $0 < L < \infty$, let $\mathcal{F}_{\uparrow,L}$ additionally have Lipschitz constant at most L. For realized feasible evidence y, define

$$\mathcal{I}_{\mathcal{G}}(y) = \{\theta_N(f) : f \in \mathcal{G},\ \theta_n(f) = y_n,\ n \le m\}. \tag{23}$$

Its exact endpoint programs are in SI Appendix, Section S6. The design frontier is instead the largest compatible width over feasible evidence:

$$\begin{aligned} \Delta_{m,N}^{\mathcal{G}} &= \sup_{\mathcal{I}_{\mathcal{G}}(y) \ne \varnothing} \operatorname{diam} \mathcal{I}_{\mathcal{G}}(y) \\ &= \sup_{\substack{f,g \in \mathcal{G} \\ \theta_n(f) = \theta_n(g)\ (n \le m)}} \big|\theta_N(f) - \theta_N(g)\big|. \end{aligned} \tag{24}$$

For $a = (a_1, \ldots, a_m) \in \mathbb{R}^m$, put

$$z_a(t) = 1 - t^N - \sum_{n=1}^{m} a_n (1 - t^n) \tag{25}$$

and define

$$H_L(z) = \sup_{\substack{c \ge 0,\ 0 \le q \le L \\ c + \int_0^1 q \le 1}} \left\{ c z(0) + \int_0^1 z(t) q(t)\, dt \right\}. \tag{26}$$

Equivalently, with $[x]_+ = \max(x, 0)$,

$$H_L(z) = \inf_{\lambda \ge 0} \left\{ \lambda + [z(0) - \lambda]_+ + L \int_0^1 [z(t) - \lambda]_+\, dt \right\}. \tag{27}$$

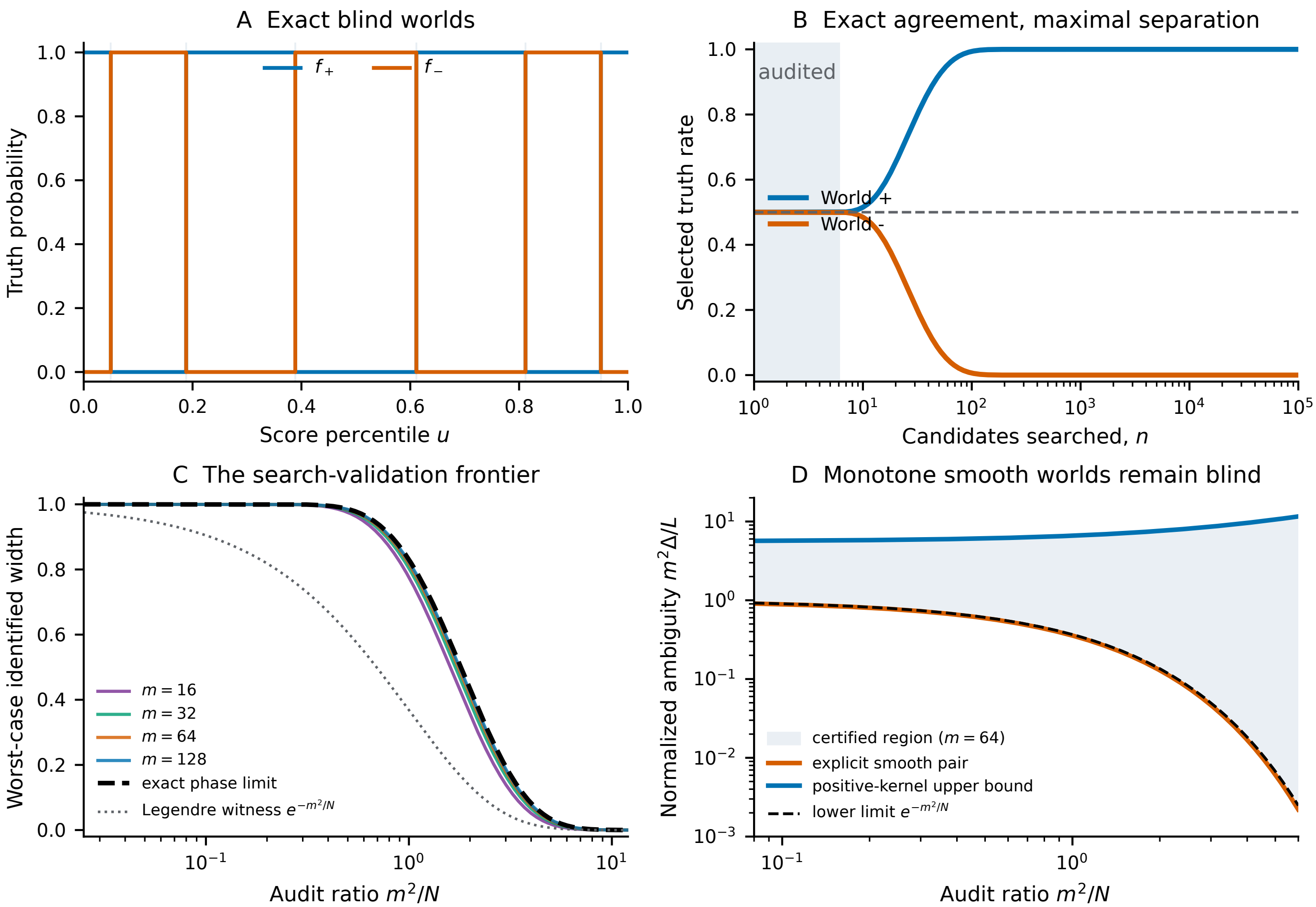

Figure 2: Exact and shape-restricted search–validation frontiers. (A) For $m = 6$, complementary Chebyshev-sign reliability surfaces produce the same first six audited reliabilities. (B) They agree throughout the audited range and then separate maximally. (C) Finite exact widths collapse under m^2/N to the Jacobi-theta frontier; the smooth Legendre witness alone understates unrestricted ambiguity. (D) Under monotonicity and a Lipschitz bound L, an explicit smooth pair and a positive-polynomial recovery bound bracket normalized ambiguity. Regularity changes its magnitude to order L/m^2 but preserves the degree-squared endpoint-resolution phase.

Theorem 3 (Exact shape-constrained frontiers)**.** *For integers $N > m \geq 1$:*

(i) For monotone reliability,

$$\Delta^{\uparrow}_{m,N} = 2 \inf_{p \in \mathcal{P}_m} \|u^N - p(u)\|_\infty. \tag{28}$$

Thus the worst monotone worlds may be chosen as step functions, and the exact validation problem is uniform monomial approximation rather than the unrestricted L^1 problem.

(ii) For every $0 < L < \infty$, monotone L-Lipschitz reliability has the exact finite dual

$$\Delta^{\uparrow,L}_{m,N} = \inf_{a \in \mathbb{R}^m} \{H_L(z_a) + H_L(-z_a)\}. \tag{29}$$

Evaluation therefore reduces to the m audit coefficients and the scalar threshold in Eq. 27.

The proof represents a monotone law as the distribution function of a subprobability measure. With a Lipschitz bound, this measure has a baseline atom and an absolutely continuous component with density capped by L. Compact-measure separation gives Eqs. 28 and 29 without a duality gap and proves attainment. Bounded-density moment theory supplies adjacent compact-set characterizations [20, Theorem 3(a–c), pp. 5–6]; the submitted object is the residual induced by this audit/deployment pair.

Corollary 2 (Constructive endpoint-resolution bounds)**.** *Let $d = \lfloor (m-1)/2 \rfloor$, let ξ_{d+1} be the largest zero on $[0,1]$ of the shifted Legendre polynomial of degree $d+1$, and put*

$$A_{m,L} = \min\left\{\frac{1}{m+1}, \frac{L}{m(m+1)}\right\}, \quad D_{m,N} = \prod_{j=1}^{m} \frac{N-j}{N+j}. \tag{30}$$

Then

$$A_{m,L} D_{m,N} \leq \Delta^{\uparrow,L}_{m,N},$$
$$\Delta^{\uparrow,L}_{m,N} \leq \min\left\{B_{m,N}, L\left(1 - \xi_{d+1} + \frac{1}{N+1}\right), 1\right\}. \tag{31}$$

If L is fixed and $m^2/N \to \tau \in (0,\infty)$, then

$$e^{-\tau} \le \liminf \frac{m^2\Delta^{\uparrow,L}_{m,N}}{L} \le \limsup \frac{m^2\Delta^{\uparrow,L}_{m,N}}{L} \le j_{0,1}^2+\tau, \quad (32)$$

where $j_{0,1} \simeq 2.4048$ is the first positive zero of J_0.

Both bounds are constructive. The lower bound builds two smooth monotone worlds from a shifted Legendre polynomial; the upper bound uses a nonnegative polynomial audit concentrated near the endpoint. Their constants do not match, so Eq. 32 is order-sharp rather than a closed form; the exact finite quantity is Eq. 29. This changes the magnitude of ambiguity to order L/m^2 while preserving the competition between m^{-2} endpoint resolution and N^{-1} deployment concentration (Fig. 2D).

The dual is numerically accessible rather than only formal. With 8,192 percentile bins, floating-point primal and continuum-dual estimates are 0.046130661 and 0.046130667 at $(m, N, L) = (4, 16, 1)$ and 0.020791486 and 0.020791501 at $(8, 64, 1)$. These are agreement diagnostics, not interval-arithmetic-certified enclosures; the rigorous constructive brackets are $[0.01409, 0.27015]$ and $[0.00449, 0.08482]$. SI Appendix, Section S6 reports convergence, optimizers, and sensitivity. Because u is percentile rank, L is invariant to monotone score transformations, but the audit prefix does not validate it. A defensible bound must come from scientific knowledge or independent candidate-level data with uniform uncertainty; otherwise the frontier should be reported over L.

From geometry to an audit decision

The population frontier determines whether a finite audit is worth extending by replication. Let $Z_{i,n} \in [0,1]$ be the width-n outcome on task i, with tasks independent and identically distributed and $\mathbb{E}Z_{i,n} = \theta_n$. For deterministic coefficients $a \in \mathbb{R}^m$ chosen independently of these audit outcomes, put $s_a = \sum_{n=1}^{m} a_n k_n$ and $\widehat{\theta}_N(a) = 1/2 + \sum_n a_n(\overline{Z}_n - 1/2)$.

Corollary 3 (Finite-task audit planner)**.** *For every a and $0 < \alpha < 1$, with probability at least $1-\alpha$,*

$$|\widehat{\theta}_N(a) - \theta_N| \le \frac{1}{2}\|k_N - s_a\|_1 + \|a\|_1\sqrt{\frac{\log(2/\alpha)}{2T}}. \quad (33)$$

The smallest right side over a prespecified coefficient class is a computable bias–noise certificate. As $T \to \infty$, twice its infimum tends to $B_{m,N}$. Same-sample outcome-dependent coefficient choice requires sample splitting or uniform concentration over the searched class.

This yields a two-gate rule. The *structural gate* asks whether $B_{m,N} \le \varepsilon$; if it fails, no number of additional independent tasks and no randomized or sequential schedule confined to widths at most m can achieve full width ε. The *sampling gate* is evaluated only after structural breadth is adequate; it asks whether the coefficient-amplified task term in Eq. 33 is small enough. Failure there can be repaired by more independent tasks. A deployment-matched intervention changes the first gate by adding a new kernel direction rather than merely reducing noise.

For $N = 100$ and 95% confidence, full width 0.10 requires at least $m = 19$ before sampling error is considered. If direct deployment-winner labels are feasible, their structural term is zero and Hoeffding's inequality gives the separate sufficient count of 738 independent tasks. When audit and deployment kernels are estimated rather than known, SI Appendix, Section S7.2 adds the explicit penalty $\delta_N + \sum_n |a_n|\delta_n$. SI Appendix, Section S3.3 gives the planning table. These calculations convert the frontier into a choice among three noninterchangeable actions: widen the audited search range, increase independent task count, or collect labels in the deployment direction.

Retrospective illustrations of the same geometry

The empirical analyses ask whether changing search width changes the truth target in objective data, whether reversals can coexist with favorable mean scaling, and whether a realistic audited prefix admits widely separated deployment values. They are mechanism demonstrations in fixed public populations, not prevalence estimates or prospective interventions; SI Appendix, Sections S8–S10, records the inferential boundary. The sampling idealization is conditional on task: task t has its own randomized percentile and reliability law f_t, with $\theta_{n,t} = \int k_n f_t$. Equal-weight macro averaging gives $\overline{\theta}_n = \int k_n \overline{f}$ for $\overline{f}$ equal to the average task law, without treating candidates from different tasks as exchangeable. Every reported best-of-n value is the exact with-replacement maximum-score functional of a fixed task-specific empirical population. The two domains vary output type, verifier construction, truth mechanism, and search range while retaining this iid score-and-select estimand. Treating problems rather than candidate outputs as the independent units, the first analysis uses the two pinned Monkey Business files [21]. Each file contains exactly 127 GSM8K problem records with 10,000 independently generated solutions and exact final-answer correctness labels; all 127 records were included for each model, with no outcome- or completeness-based exclusion (2.54 million candidates in total).

For each problem and model, a seeded 2,000-solution reference split estimated the frequency of each normalized final answer. That frequency served as a simple consensus score. A disjoint 8,000-solution deployment split supplied score ranks and truth labels. We evaluated the empirical best-of-n law exactly for $n = 1, 2, 4, \ldots, 4096$, averaging uniformly across tied winning scores. Problem-bootstrap intervals used 20,000 resamples. Ten independent reference/deployment splits assessed stability.

Consensus ranking was strong on average. Macro within-problem AUC was 0.907 (95% bootstrap interval, 0.862–0.947) over the 125 of 127 mixed-label 8B deployment populations, and 0.962 (0.923–0.990) over the 107 of 127 mixed-label 70B populations; AUC is undefined for a single-class problem. Mean selected truth rose from 0.768 at $n = 1$ to 0.882 at $n = 4096$ for 8B, and from 0.931 to 0.968 for 70B (Fig. 3A). Yet 14 of 127 8B problems and three 70B problems lost more than one percentage point; worst losses were 0.480 and 0.360 (Fig. 3C). Across 10 splits, the 8B harm count stayed between 13 and 15 and the same three 70B problems were harmed in every split (Fig. 3E). The data do not imply that the extremal Chebyshev worlds are typical. They establish that changing the search kernel changes a practically relevant reliability target even when aggregate ranking looks favorable.

The independent replication uses 164 HumanEval+ tasks from the CodeRM release [22]. For each task and each Llama-3 model, we analyzed 100 candidate programs. The verifier score is the fraction of 100 CodeRM-8B-generated unit tests passed; correctness is the separate HumanEval+ `plus_status` label. The task-level analysis therefore contains 32,800 programs and 3.28 million candidate–unit-test executions, but inference remains across 164 tasks. Exact tie-averaged best-of-n curves and 20,000 task bootstraps used the same estimand as the math experiment. All 164 tasks contained both truth classes for each model and therefore entered the macro AUCs: 0.931 (0.898–0.959) for 8B and 0.881 (0.818–0.938) for 70B. Mean selected truth rose from 0.536 to 0.715 and from 0.737 to 0.788 between $n = 1$ and 100 (Fig. 3B). Nevertheless, 10 and four tasks lost more than one point, with worst losses of 0.722 and 0.600 (Fig. 3D). The reversal phenomenon therefore survives a change of task, verifier, ground-truth mechanism, and search range.

A retrospective empirical compatibility construction. We next asked the same question of the CodeRM data that the theorem asks of an unknown system. Within each task, tie-group truth was spread uniformly over its score-rank intervals; averaging the 164 task laws reproduces every macro best-of-n mean exactly. We then concealed the best-of-100 value, retained only the plug-in audits $\widehat{\theta}_1, \ldots, \widehat{\theta}_8$, and optimized θ_{100} within the subclass of bounded rank–truth laws that are constant on 1,000 percentile bins.

For 8B, the observed best-of-100 truth was 0.715 (95% task-bootstrap interval, 0.650–0.778), while two such laws matching all eight plug-in audits reached 0.186 and 0.992 at deployment (SI Appendix, Fig. S1A,C). For 70B, the observed value was 0.788 (0.725–0.847), while compatible witnesses reached 0.305 and 0.996 (SI Appendix, Fig. S1B,D). Because every optimizer is a bounded continuum step function, these separations are genuine feasible witnesses and therefore lower bounds on the unrestricted continuum identified width; the displayed endpoints are not claimed to solve that unrestricted problem exactly. Orthogonalizing the same equality row space reduced raw audit residuals below 1.3×10^{-14}, and changing the grid from 200 to 1,000 bins moved every displayed endpoint by less than 0.007. Allowing 95% simultaneous max-t task-bootstrap bands produced 1,000-bin feasible spans $[0.003, 1.000]$ and $[0.016, 1.000]$. At $m = 20$, the corresponding plug-in spans narrowed to 0.047 and 0.038, yet the finite-task band spans remained 0.897 and 0.803. This is the two-gate distinction in Corollary 3: broader interventions narrow the structural witness, whereas estimated high-order moments can still amplify task noise. SI Appendix, Section S10, reports the construction and values for $m = 2$ through 20.

A retrospective masked-label acquisition replay. To examine the design consequence within the fixed CodeRM population, we treated every candidate truth as hidden until selected by a score-only audit rule. Each draw sampled a task uniformly and then a uniform candidate, a best-of-8 winner, a randomized top-5% score percentile, or a best-of-100 deployment winner. Horvitz–Thompson estimates targeted the macro best-of-100 truth. Across 5,000 replayed acquisitions at each budget, 500 uniform labels left 95th-percentile absolute errors of 0.184 and 0.110 for 8B and 70B; top-tail labels reduced them to 0.047 and 0.038, and deployment-winner labels to 0.039 and 0.036 (Fig. 3F). The same top-5% rule, budgets, and estimator were used for both models without model-specific retuning. Its observed truncation biases were below 10^{-4}; its distribution-free omitted-mass bound was $0.95^{100} = 0.0059$. SI Appendix, Section S9, reports the full budget curves, estimator, and the earlier asymptotic comparison. This is a within-pool estimator comparison on public labels, not prospective evidence about intervention effectiveness or realized costs.

A frozen-rule held-out test. We then separated design choice from evaluation. A seeded, label-blind split assigned the 164 shared CodeRM tasks to 82 discovery and 82 held-out tasks. Using discovery outcomes only, we chose among prespecified 5%, 10%, and 20% score-tail audits by mean 95th-percentile error across the two models; all three leave at most $0.95^{100} = 0.0059$ of target mass unsupported. The 5% rule was selected and frozen before either held-out target or error was computed. At 500 labels and 5,000 replays, its held-out 95th-percentile errors were 0.049 rather than 0.180 under uniform labels for 8B, and 0.035 rather than 0.076 for 70B. Held-out absolute biases were below 8×10^{-5}. This result shows that the audit-design choice transferred across a task split within CodeRM; because candidates and labels were already public, it is still retrospective and does not estimate prospective effect or acquisition cost.

Table 1 makes the design implication concrete on one fixed target. Extending the prefix from $m = 8$ to 20 sharply narrows the 1,000-bin plug-in witness, yet finite-task feasible spans remain wide because high-order moment estimation amplifies noise. Holding the label budget at 500 and moving acquisition toward the deployment tail instead reduces held-out replay error by factors 3.67 and 2.15 relative to uniform labels. The metrics are not interchangeable and do not establish a causal ranking of audit programs; together they show why "collect more labels" is incomplete without specifying audit breadth, independent units, and the direction in which labels are collected.

The masked-label result is not a theorem about all models, and importance weighting need not be the preferred estimator. Within this fixed public population, it provides a finite, unit-matched demonstration of the span-design principle: collecting labels in the deployment direction can dominate replicating labels under a mismatched intervention. The held-out split strengthens the inference from in-sample feasibility to transfer across tasks within CodeRM; it does not establish prospective effectiveness. The complete tables, formulas, split assignments, CodeRM replication, and recon-

Table 1: What changes when the audit follows the best-of-100 target. The first two rows report widths of 1,000-bin CodeRM feasible constructions under simultaneous task bands; they are witnessed lower bounds on unrestricted continuum ambiguity. The last two report held-out 95th-percentile absolute errors across 5,000 masked-label replays at 500 labels after the tail fraction was selected on a separate 82-task discovery split. These are different metrics, shown together to separate structural breadth, task precision, and label direction. The comparison is retrospective, not a prospective cost-effectiveness claim.

Evidence collected	Structural or sampling consequence	8B	70B
Mean prefix, $n \leq 8$	$B_{8,100} = 0.906$; constructed task-band span	0.997	0.984
Mean prefix, $n \leq 20$	$B_{20,100} = 0.057$; constructed task-band span	0.897	0.803
500 held-out uniform labels	Unbiased importance weighting; high weight variance	0.180	0.076
500 held-out frozen top-5% labels	Tail-matched; omitted target mass ≤ 0.0059	0.049	0.035

struction procedures are in SI Appendix, Sections S8–S10.

Discussion

AI search changes the target that validation must certify. The span theorem expresses the resulting geometry as the organizing separation

$$\text{certificate width} = \text{structural blindness} + \text{sampling uncertainty}.$$

The finite best-of-N theorem makes the first term exact: reliability through width m cannot certify deployment beyond Eq. 14, and the same lower bound applies to every randomized or sequential reliability-mean schedule confined to widths at most m. This is an audit-complexity limit rather than a defect of one prefix protocol. Theorem 3 shows that the interpretation is not confined to oscillatory Chebyshev worlds: regularity changes the magnitude, but the competition between m^{-2} audit resolution and N^{-1} deployment concentration remains. More data inside the same span attack precision but cannot beat its structural floor. A thousand evaluations produced inside one epistemic frame may therefore constitute one validation intervention.

This changes evaluation practice. The allowable prompts, search budgets, tools, subpopulations, and workflows should be stated as a deployment family. Uncertainty should cover that family simultaneously unless selection is demonstrably restricted or stable. New tests should be chosen for span expansion: ground-truthing winners, near-winners, or perturbed workflows can be more informative than labeling more random candidates. In CodeRM, the discovery-selected tail rule retained its advantage after it was frozen and moved to held-out tasks, a finite check that the design choice was not selected on the evaluation tasks themselves. Any smoothness, exchangeability, or tail assumption doing the extrapolation should be visible and stress-tested.

The span theorem requires a stable reliability surface and known or independently estimated kernels. The closed form further requires iid candidates, scalar ranking, randomized ties, maximum selection, and bounded truth; adaptive generation is outside $k_N(u) = Nu^{N-1}$ unless its history and policy are modeled. The empirical work is equally delimited: its independent units are 127 problems and 164 tasks, not millions of within-task outputs or test executions. The compatible worlds establish feasibility conditional on the prefix, not probability or prevalence. The held-out split blocks outcome-guided design selection on the evaluation tasks, but the replay still uses existing public candidates and labels rather than a prospective randomized intervention. Thus the data provide construct-level replication and retrospective held-out mechanism evidence only within iid score-and-select. A third domain could extend external validity but would not verify a distribution-free theorem. Universal guarantees are intentionally conservative; fixed, independent, or stable selection may permit narrower inference.

The framework is therefore not an argument against AI search. Search improved mean accuracy in both public experiments. It is an argument that search changes what must be validated. When AI expands the set of selectable decisions faster than evaluation expands the span of interventions, apparent precision can grow while scientific identification deteriorates.

Code and data availability

All manuscript source, analysis code, processed data, theorem checks, optimization outputs, and deterministic figure code are available at `https://github.com/rfitas-lab/geometry-of-ai-validation`. Monkey Business generations and labels are available at `https://huggingface.co/datasets/ScalingIntelligence/monkey_business`; CodeRM programs, annotations, and execution outputs are linked from `https://github.com/RUCKBReasoning/CodeRM`. Raw generations, programs, and execution logs are not redistributed because of their size.

Competing interests. The author declares no competing interest.

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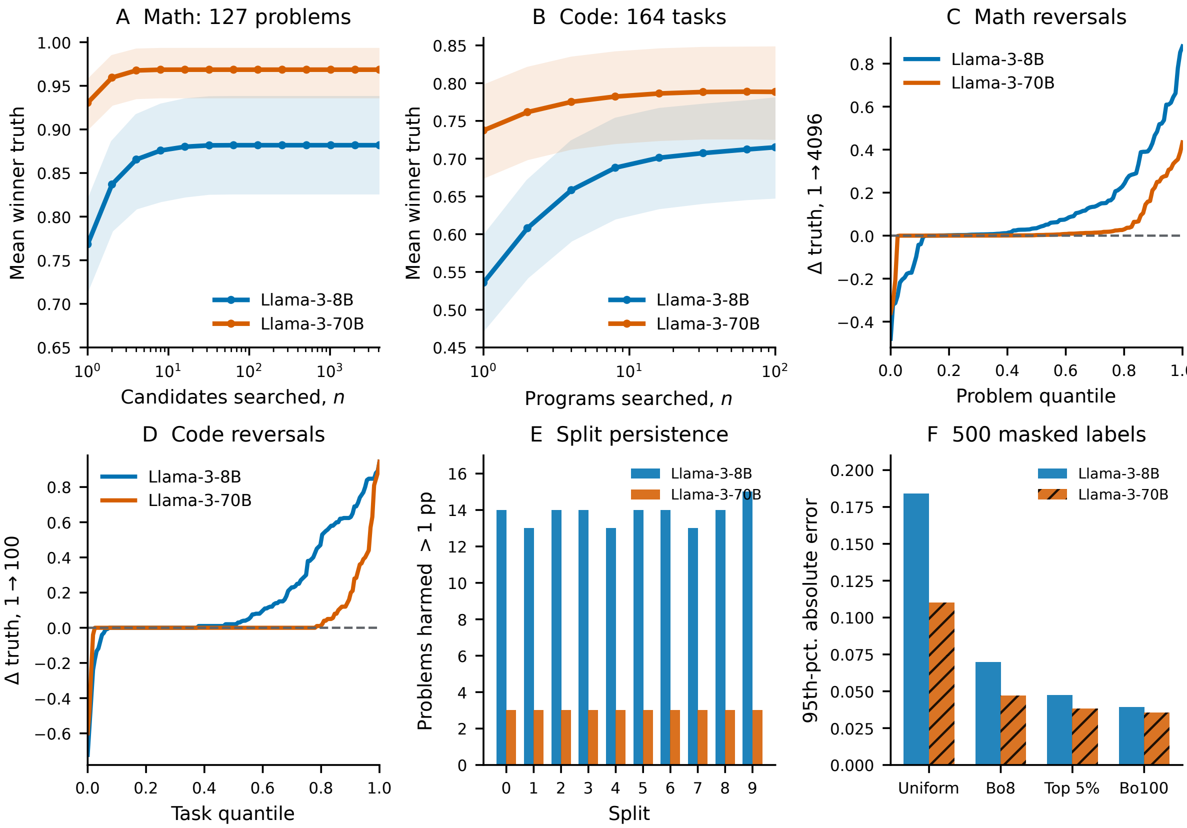

Figure 3: Search changes reliability in two objective domains. (A) Mean selected truth and 95% problem-bootstrap intervals across 127 GSM8K problems (2.54 million solutions) under answer-consensus ranking. (B) Independent replication across 164 HumanEval+ tasks (32,800 programs; 3.28 million CodeRM unit-test executions). (C,D) Sorted task-level changes show severe reversals despite favorable means and high macro AUC. (E) GSM8K harm counts persist across 10 disjoint reference/deployment splits; macro AUC remains 0.90–0.96. (F) In 5,000 retrospective masked-label replays at a 500-label budget, score-matched CodeRM audits reduce the 95th percentile of absolute best-of-100 estimation error relative to uniform candidate labels. Truth is used only when a candidate is sampled; full labels score the emulation afterward.

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Supporting Information

The geometry of AI validation: Exact certification limits for iid best-of-N search

Ricardo Fitas

This appendix gives complete proofs of the span theorem, post-selection and noisy-capacity results, exact unrestricted certification frontier, reliability-mean information-maximality result, evidence-specific shape programs, exact worst-evidence shape duals, constructive endpoint bounds, finite-task planner, and kernel-perturbation result. It derives the scalar-threshold capped-tail algorithm, Jacobi-theta and fixed-Lipschitz phase laws, reports numerical and Lipschitz-sensitivity checks, states scope conditions, documents retrospective analyses across 127 mathematical-reasoning problems and 164 code tasks, and records a concise contribution boundary.

S1. Exact inverse geometry of validation

S1.1. Setup and notation

Let $(\mathcal{X}, \mathcal{A}, \mu)$ be a σ-finite measure space. For $g \in L^1(\mu)$ and $f \in L^\infty(\mu)$, write

$$\langle g, f\rangle = \int_{\mathcal{X}} g(x) f(x)\, d\mu(x).$$

The admissible reliability surfaces are

$$\mathcal{F} = \{f \in L^\infty(\mu) : 0 \le f \le 1 \ \mu\text{-almost everywhere}\}. \tag{S1}$$

Validation kernels $v_1, \dots, v_p \in L^1(\mu)$ define the operator

$$Kf = (\langle v_1, f\rangle, \dots, \langle v_p, f\rangle). \tag{S2}$$

Let $S_V = \operatorname{span}\{v_1, \dots, v_p\}$. Because S_V is finite-dimensional, it is closed in L^1. A target kernel $k \in L^1$ defines $R_k(f) = \langle k, f\rangle$.

The following elementary identity is what makes the unit interval constraint exact rather than merely convenient.

Lemma S1 (Difference set)**.** *The set of all differences of two reliability surfaces is the L^∞ unit ball:*

$$\{f - g : f, g \in \mathcal{F}\} = \{h \in L^\infty : \|h\|_\infty \le 1\}. \tag{S3}$$

Proof. Every difference $f - g$ lies between -1 and 1. Conversely, for any h in the unit ball, $f = (1 + h)/2$ and $g = (1 - h)/2$ both belong to $\mathcal{F}$ and satisfy $f - g = h$. □

S1.2. Proof of the span identity

The annihilator of S_V is

$$S_V^\perp = \{h \in L^\infty : \langle s, h\rangle = 0 \text{ for every } s \in S_V\}. \tag{S4}$$

Standard quotient-space duality gives the following exact recovery identity; its statistical optimal-recovery antecedent is discussed by Donoho [4, Secs. 2–3, pp. 242–259]:

$$\|k + S_V\|_{L^1/S_V} = \inf_{s \in S_V} \|k - s\|_1 = \sup_{\substack{h \in S_V^\perp \\ \|h\|_\infty \le 1}} |\langle k, h\rangle|. \tag{S5}$$

The supremum is attained: apply Hahn–Banach to the one-dimensional span of the coset $k + S_V$ in L^1/S_V, and represent the resulting bounded functional by an element of L^∞.

If $f, g \in \mathcal{F}$ satisfy $Kf = Kg$, then $h = f - g$ belongs to $S_V^\perp$ and has $\|h\|_\infty \le 1$. Lemma S1 also shows that every such h is a difference of admissible f, g. Therefore

$$\begin{aligned} \sup_{\substack{f,g \in \mathcal{F} \\ Kf = Kg}} |R_k(f) - R_k(g)| &= \sup_{\substack{h \in S_V^\perp \\ \|h\|_\infty \le 1}} |\langle k, h\rangle| \\ &= \inf_{s \in S_V} \|k - s\|_1. \end{aligned} \tag{S6}$$

This proves the sharp-pair-ambiguity statement.

If $k \in S_V$, write $k = \sum_j a_j v_j$. Then $R_k(f) = a^\top K f$, so the target is identified. Conversely, if $k \notin S_V$, the right side of Eq. S6 is positive, and two validation-equivalent worlds have different target values. This proves the if-and-only-if statement.

Assume now that every v_j and k integrates to one. The constant reliability surface $f_0 = 1/2$ produces the central evidence $K f_0 = (1/2, \ldots, 1/2)$. Every surface with that evidence can be written as $f = (1+h)/2$, where $h \in S_V^\perp$ and $\|h\|_\infty \le 1$. Hence

$$R_k(f) = \frac{1}{2} + \frac{1}{2}\langle k, h\rangle. \tag{S7}$$

Equation S5 shows that the supremum and infimum are $(1 \pm b(k;V))/2$. Convex scaling th, $0 \le t \le 1$, fills every intermediate value. Thus the central identified set is exactly

$$\left[\frac{1-b(k;V)}{2}, \frac{1+b(k;V)}{2}\right]. \tag{S8}$$

This completes the proof of the span identity stated in the main text. □

S1.3. Interpretation and variants

For noncentral evidence, the identified interval can be narrower because the box constraints $0 \le f \le 1$ may bind asymmetrically. Equation S6 is the largest width over all feasible evidence, and Eq. S8 exhibits evidence at which it is attained. For an infinite collection of validation kernels, S_V is replaced by its L^1 closure. Signed kernels are allowed; if kernels are not normalized, the same proof holds with the appropriate baseline integrals in Eq. S7.

The L^1 metric is forced by the model class rather than selected for convenience: the reliability perturbation is bounded in L^∞, whose dual pairing produces L^1 distance. If the admissible surface were restricted by another norm or shape class, the corresponding support function would replace Eq. S5.

S2. Post-evaluation target selection and simultaneous protection

S2.1. Exact equivalence for a finite family

Let $\mathcal{W} = \{1, \ldots, M\}$ and let $C_w(Y)$ be measurable random confidence sets. Fix a reliability surface f. Define the simultaneous-coverage event

$$A_f(Y) = \bigcap_{w=1}^{M} \{R_w(f) \in C_w(Y)\}. \tag{S9}$$

On A_f^c, let $\widehat{w}_f(Y)$ be the smallest index whose confidence set misses $R_w(f)$; on A_f, set $\widehat{w}_f(Y) = 1$. Then

$$\{R_{\widehat{w}_f}(f) \in C_{\widehat{w}_f}(Y)\} = A_f. \tag{S10}$$

Consequently, if selected-target coverage is at least $1-\alpha$ for every measurable selector, it is at least $1-\alpha$ for $\widehat{w}_f$, which implies simultaneous coverage. The converse is immediate because simultaneous coverage implies coverage for any selected coordinate.

The selector constructed for the proof depends on the fixed data-generating f. That is legitimate because a universal guarantee quantifies over every selector and every f. It does not imply that an analyst knows f or can implement the first-failure selector. It shows that no procedure can advertise validity under arbitrary unknown selection while avoiding the simultaneous event. For countable families the same argument uses the first failing index. For uncountable families, measurability and separability conditions are needed; one may work with a countable dense subclass when both target and confidence processes are sample-continuous.

S2.2. Gaussian selected-target example

Let $Y_w \overset{\text{iid}}{\sim} N(0,1)$ and $z = \Phi^{-1}(0.975)$. Each interval $[Y_w - z, Y_w + z]$ covers its fixed target zero with probability 0.95. If $\widehat{w} = \arg\max_w Y_w$, selected coverage is

$$\begin{aligned}\mathbb{P}\{0 \in [Y_{\widehat{w}} - z, Y_{\widehat{w}} + z]\} &= \mathbb{P}\{-z \le \max_w Y_w \le z\} \\ &= \Phi(z)^M - \Phi(-z)^M.\end{aligned} \tag{S11}$$

The two-sided simultaneous critical value c_M solves

$$\{2\Phi(c_M) - 1\}^M = 0.95, \qquad c_M = \Phi^{-1}\left[\frac{1 + 0.95^{1/M}}{2}\right]. \tag{S12}$$

The exact values in the machine-readable table are produced by `code/reproduce_geometry_validation.py`. The example isolates stochastic post-selection. It has no structural blind residual because each coordinate is observed directly.

Table S1: Selection destroys pointwise Gaussian coverage. Coverage uses the ordinary 95% interval after selecting the largest of M independent estimates. Width inflation is $c_M/1.96$.

M	Selected coverage	Simultaneous width inflation
1	0.950	1.000
5	0.881	1.311
10	0.776	1.428
20	0.603	1.539
50	0.282	1.675
100	0.080	1.772
1,000	1.01×10^{-11}	2.066
100,000	$< 10^{-300}$	2.562

S3. Noisy validation capacity and intervention design

S3.1. A simultaneous bias–noise certificate

Suppose

$$Y_j = \langle v_j, f\rangle + n^{-1/2}\xi_j, \qquad \xi \sim N(0, \Sigma). \tag{S13}$$

For every target w, choose $a_w \in \mathbb{R}^p$, define $s_w = \sum_j a_{wj} v_j$, and set

$$\widehat{R}_w = a_w^\top Y + \frac{1}{2}\int (k_w - s_w)\, d\mu. \tag{S14}$$

Then

$$\widehat{R}_w - R_w(f) = n^{-1/2} a_w^\top \xi - \left\langle k_w - s_w, f - \frac{1}{2}\right\rangle, \tag{S15}$$

and Hölder's inequality gives

$$\left|\left\langle k_w - s_w, f - \frac{1}{2}\right\rangle\right| \le \frac{1}{2}\|k_w - s_w\|_1. \tag{S16}$$

Let

$$q_{1-\alpha}(A, \Sigma) = \inf\left\{q : \mathbb{P}\left(\sup_w |a_w^\top \xi| \le q\right) \ge 1-\alpha\right\}. \tag{S17}$$

Combining Eqs. S15–S17 proves, simultaneously over all targets and all $f \in \mathcal{F}$,

$$|\widehat{R}_w - R_w(f)| \le \frac{1}{2}\|k_w - s_w\|_1 + \frac{q_{1-\alpha}(A,\Sigma)}{\sqrt{n}} \tag{S18}$$

with probability at least $1-\alpha$.

For a finite family of M targets, if $a_w^\top \Sigma a_w \le \sigma_A^2$ for every w, a union bound gives the explicit sufficient value

$$q_{1-\alpha}(A, \Sigma) \le \sigma_A \sqrt{2\log(2M/\alpha)}. \tag{S19}$$

More sharply, let

$$G(A, \Sigma) = \mathbb{E}\sup_w |a_w^\top \xi| \tag{S20}$$

be the Gaussian width of the symmetrized coefficient image. Gaussian concentration implies

$$q_{1-\alpha}(A, \Sigma) \le G(A, \Sigma) + \sigma_A\sqrt{2\log(1/\alpha)}. \tag{S21}$$

The displayed bounds make the two geometries explicit: L^1 approximation controls worst-case structural bias, while Euclidean/Gaussian geometry controls noise amplification.

One may optimize the common half-width over coefficient fields:

$$C_{n,\alpha}(\mathcal{K}; V, \Sigma) = \inf_{\{a_w\}}\left\{\frac{1}{2}\sup_w \left\|k_w - \sum_j a_{wj} v_j\right\|_1 + \frac{q_{1-\alpha}(A,\Sigma)}{\sqrt{n}}\right\}. \tag{S22}$$

This is a constructive upper bound for universal confidence-band width. As $n \to \infty$, the stochastic term disappears and the infimum of the first term is $B(\mathcal{K}; V)/2$. Equation S6 shows that no honest central-evidence band can have a smaller worst-case asymptotic half-width. At finite n, minimizing the structural and stochastic terms separately need not be optimal: coefficients that approximate a target closely may amplify noise.

S3.2. Adding realizable interventions

Let $\mathcal{D} \subset L^1(\mu)$ be the set of feasible new intervention kernels. After choosing q of them, the smallest remaining structural width is

$$\mathfrak{d}_q(\mathcal{K} \mid V, \mathcal{D}) = \inf_{d_1,\ldots,d_q \in \mathcal{D}} \sup_{k \in \mathcal{K}} \operatorname{dist}_1\{k, \operatorname{span}(S_V, d_1, \ldots, d_q)\}. \tag{S23}$$

The sequence is nonincreasing in q and equals zero when the augmented span contains every target kernel. If $\mathcal{D}$ permits every q-dimensional extension, Eq. S23 is a relative Kolmogorov-width problem [23]. Real validation design is more constrained: a signed or oscillatory mathematical direction may not correspond to a population that can be sampled or a workflow that can be executed. The dictionary makes that constraint explicit.

When interventions have different costs or noise covariance, Eq. S23 should be replaced by the finite-n objective in Eq. S22. The core qualitative distinction survives: replication changes the noise term, whereas a new kernel can change the closed span.

S3.3. A distribution-free finite-task planner

The Gaussian model above isolates covariance geometry. For the best-of-N application, a fully explicit task-level certificate needs only bounded outcomes. Let $Z_i = (Z_{i,1}, \ldots, Z_{i,m}) \in [0,1]^m$ be independent and identically distributed audit vectors across T tasks, with $\mathbb{E}Z_{i,n} = \theta_n$. Dependence among widths within the same task is unrestricted. Fix coefficients $a \in \mathbb{R}^m$ independently of the observed audit outcomes and define

$$s_a = \sum_{n=1}^{m} a_n k_n, \qquad \widehat{\theta}_N(a) = \frac{1}{2} + \sum_{n=1}^{m} a_n \left(\overline{Z}_n - \frac{1}{2}\right). \tag{S24}$$

The same centering identity as Eq. S15 gives

$$\widehat{\theta}_N(a) - \theta_N = \sum_{n=1}^{m} a_n(\overline{Z}_n - \theta_n) - \left\langle k_N - s_a, f - \frac{1}{2}\right\rangle. \tag{S25}$$

The deterministic term is at most $\|k_N - s_a\|_1/2$. Across tasks, the scalar $a^\top Z_i$ has range length at most $\|a\|_1$, regardless of dependence among its coordinates. Hoeffding's inequality therefore gives

$$\mathbb{P}\left\{\left|\sum_{n=1}^{m} a_n(\overline{Z}_n - \theta_n)\right| > \|a\|_1 \sqrt{\frac{\log(2/\alpha)}{2T}}\right\} \le \alpha. \tag{S26}$$

Combining the two terms gives the following standalone SI finite-task certificate. The computable distribution-free half-width is

$$C_{T,\alpha}^{(m,N)} = \inf_{a \in \mathbb{R}^m} \left\{\frac{1}{2}\left\|k_N - \sum_{n=1}^{m} a_n k_n\right\|_1 + \|a\|_1 \sqrt{\frac{\log(2/\alpha)}{2T}}\right\}. \tag{S27}$$

This is a convex bias–noise tradeoff over outcome-independent coefficients, which may be selected from the known kernels, T, and α. Choosing the L^1-best approximant at finite T need not be optimal because its coefficients may amplify noise. If a is chosen after inspecting the same outcomes, Eq. S26 does not by itself protect that choice; one needs sample splitting or a uniform concentration bound over the searched coefficient class. As $T \to \infty$, twice Eq. S27 decreases to $B_{m,N}$. Hence $B_{m,N} \le \varepsilon$ is a necessary infinite-task gate for any full-width target ε based only on the prefix.

If a new audit samples one best-of-N winner per independent task, set the new intervention equal to k_N. The structural term is zero, the coefficient norm is one, and a full $1-\alpha$ interval of width at most ε is guaranteed whenever

$$T \ge \frac{2\log(2/\alpha)}{\varepsilon^2}. \tag{S28}$$

Table S2 keeps this sufficient direct-audit calculation separate from the necessary prefix breadth. The prefix column does not assert that its listed m is sufficient at finite T; remaining sampling error must still fit inside the width budget through Eq. S27 or a covariance-adaptive analogue.

S4. Exact best-of-N frontier

S4.1. Rank normalization and winner kernels

Let (S,T) be a candidate score and a bounded outcome $T \in [0,1]$. If S has a continuous distribution function F_S, then $U = F_S(S)$ is uniform on $[0,1]$. With score atoms, define the randomized percentile

$$U = F_S(S^-) + Z\{F_S(S) - F_S(S^-)\}, \qquad Z \sim \operatorname{Unif}(0,1), \tag{S29}$$

where Z is independent. Uniform tie breaking under the original score is equivalent to maximizing U. For an atom whose score mass occupies a percentile interval I, the independent jitter implies that $\mathbb{E}(T \mid U = u)$ is constant for almost every

Table S2: Finite certificate planning for $N = 100$ and 95% confidence. Minimum m is necessary even with infinitely many tasks. The direct-winner task count is a separate sufficient Hoeffding design. Exact values are in `results/certificate_planning.csv`.

Desired full width	Minimum m	$B_{m,100}$	Direct-winner tasks
0.50	13	0.446	30
0.25	16	0.213	119
0.20	17	0.159	185
0.10	19	0.082	738
0.05	21	0.039	2,952

$u \in I$. Thus a known atomic score distribution adds interval-constancy restrictions to the reliability class. The unrestricted frontier below ranges over the stated model class and remains exact because its extremal joint laws may use a continuous score. Conditioning on a particular atomic score law can only narrow that frontier; the empirical programs in Sections S8–S10 preserve the observed tie intervals explicitly.

For n iid candidates, the maximum percentile has distribution function u^n and density

$$k_n(u) = nu^{n-1}. \tag{S30}$$

If $f(u) = \mathbb{E}(T \mid U = u)$, the selected mean is

$$\theta_n = \int_0^1 k_n(u) f(u)\, du. \tag{S31}$$

The first m kernels span $\mathcal{P}_{m-1}$ because nonzero scalar multiples do not change span.

S4.2. A Chebyshev-sign annihilator

Let $U_m(x)$ be the degree-m Chebyshev polynomial of the second kind,

$$U_m(\cos\theta) = \frac{\sin\{(m+1)\theta\}}{\sin\theta}, \tag{S32}$$

and define

$$\sigma_m(u) = \operatorname{sgn} U_m(2u-1). \tag{S33}$$

Its roots are

$$u_j = \frac{1+\cos\{j\pi/(m+1)\}}{2}, \qquad j = 1,\ldots,m. \tag{S34}$$

Lemma S2 (Exact annihilation)**.** *For every polynomial $q \in \mathcal{P}_{m-1}$,*

$$\int_0^1 q(u)\sigma_m(u)\, du = 0. \tag{S35}$$

Proof. Set $2u - 1 = \cos\theta$. Up to a constant factor, the integral is

$$\int_0^\pi q\left(\frac{1+\cos\theta}{2}\right) \sin\theta \,\operatorname{sgn}\sin\{(m+1)\theta\}\, d\theta. \tag{S36}$$

The factor $q((1+\cos\theta)/2)\sin\theta$ is a finite linear combination of $\sin(\ell\theta)$ for $1 \le \ell \le m$. In $L^2(0,\pi)$, the square wave $\operatorname{sgn}\sin\{(m+1)\theta\}$ has a sine series containing only frequencies $(2r+1)(m+1)$, $r \ge 0$. Orthogonality of the sine basis makes Eq. S36 zero. Values at the finitely many roots do not affect the integral. □

S4.3. C0: classical L^1 approximation input

The canonical points for polynomial L^1 approximation—the roots of the Chebyshev polynomial of the second kind—date to Bernstein. The canonical-point, generalized-quadrature, and orthogonal-signature framework is treated explicitly by Bojanov, Braess, and Dyn [15, pp. 335–353]; broader L^1 characterizations are reviewed by Watson [16, pp. 1–15]. The calculation below is therefore a direct specialization, not a claim to a new best-approximation theorem. This subsection ends exactly where the classical input ends: with the best approximant, its error sign, and its distance.

Let $N > m$ and let p_{m-1} be the unique polynomial of degree at most $m-1$ interpolating $k_N(u) = Nu^{N-1}$ at the nodes in Eq. S34. The divided-difference remainder is

$$k_N(u) - p_{m-1}(u) = k_N[u_1,\ldots,u_m,u] \prod_{j=1}^m (u - u_j). \tag{S37}$$

The mth derivative of k_N is positive on $(0,1]$, so the divided difference is positive almost everywhere. The product in Eq. S37 is a positive multiple of $U_m(2u-1)$. Therefore

$$\operatorname{sgn}\{k_N(u) - p_{m-1}(u)\} = \sigma_m(u) \quad \text{almost everywhere.} \tag{S38}$$

For any $q \in \mathcal{P}_{m-1}$, Lemma S2 and the subgradient inequality give

$$\begin{aligned} \|k_N - (p_{m-1} + q)\|_1 &\geq \int_0^1 \sigma_m(u)\{k_N(u) - p_{m-1}(u) - q(u)\}\, du \\ &= \int_0^1 |k_N(u) - p_{m-1}(u)|\, du. \end{aligned} \tag{S39}$$

Thus p_{m-1} is an L^1 best approximant. Moreover,

$$\operatorname{dist}_1(k_N, \mathcal{P}_{m-1}) = \int_0^1 k_N(u)\sigma_m(u)\, du, \tag{S40}$$

because both p_{m-1} and every other lower-degree polynomial are annihilated by σ_m.

To evaluate Eq. S40, integrate k_N over the alternating intervals between roots. Because $\int_0^a k_N(u)du = a^N$ and the sign is positive next to $u = 1$, telescoping gives

$$\begin{aligned} B_{m,N} &= 1 + 2\sum_{r=1}^{m}(-1)^r \left[\frac{1 + \cos\{r\pi/(m+1)\}}{2}\right]^N \\ &= 1 + 2\sum_{r=1}^{m}(-1)^r \cos^{2N}\left\{\frac{r\pi}{2(m+1)}\right\}. \end{aligned} \tag{S41}$$

This proves the distance formula.

S4.4. V1–V2: bounded-reliability certification consequence

The statistical object begins here. The approximation result in Section S4.3 does not by itself specify a feasible bounded reliability model, a common audit vector, an identified set, or a minimax statement over such sets. Those are the additional claims proved next.

Define complete binary joint laws $P_\pm$ by $U_\pm \sim \text{Unif}(0,1)$ and $T_\pm \mid U_\pm = u \sim \text{Bernoulli}\{f_\pm(u)\}$, where $f_\pm = (1 \pm \sigma_m)/2$, and take candidate sequences iid under each law. For every $n \leq m$, $k_n \in \mathcal{P}_{m-1}$, so Lemma S2 gives

$$\theta_n^+ - \theta_n^- = \int_0^1 k_n\sigma_m = 0, \qquad \theta_n^+ = \theta_n^- = \frac{1}{2}. \tag{S42}$$

At N, Eq. S40 gives

$$\theta_N^\pm = \frac{1 \pm B_{m,N}}{2}. \tag{S43}$$

For every $\lambda \in [0,1]$, the joint law with $U \sim \text{Unif}(0,1)$ and $T \mid U = u \sim \text{Bernoulli}\{\lambda f_+(u) + (1-\lambda) f_-(u)\}$ is again a valid iid candidate world, has every audited reliability equal to $1/2$, and fills the entire interval between the two deployment endpoints. The exact span identity says no validation-equivalent pair can be separated by more than the L^1 distance. Thus the central identified set is exactly $[(1 - B_{m,N})/2, (1 + B_{m,N})/2]$, not merely a pair of separated witnesses. If $N \leq m$, k_N already lies in $\mathcal{P}_{m-1}$ and the distance is zero. □

S4.5. Why Chebyshev improves the orthogonal-polynomial witness

A shifted Legendre polynomial L_m also annihilates $\mathcal{P}_{m-1}$ and satisfies $\|L_m\|_\infty \leq 1$. It produces the valid gap

$$D_{m,N} = N\int_0^1 u^{N-1}L_m(u)\, du = \prod_{j=1}^{m}\frac{N-j}{N+j}. \tag{S44}$$

Therefore $D_{m,N} \leq B_{m,N}$. The Legendre construction is smooth and convenient, but it is not generally extremal for the L^1/L^∞ geometry. The exact dual extremizer is the discontinuous sign of U_m, and the exact primal solution interpolates at the roots of U_m. This distinction converts a lower-bound scale into a solved minimax frontier.

S4.6. Optimality among low-width reliability-mean audits

Let an admissible audit kernel be any randomized mixture of best-of-n winner kernels with $n \leq m$,

$$a_\ell(u) = \sum_{n=1}^{m} q_{\ell n}k_n(u), \qquad q_{\ell n} \geq 0, \quad \sum_{n=1}^{m} q_{\ell n} = 1. \tag{S45}$$

If the budget realization is retained, the audit reveals a subset or linear combination of the individual θ_n; if it is not retained, it reveals the mixture mean $\langle a_\ell, f\rangle$. In either case every revealed kernel lies in $\mathcal{P}_{m-1}$. For a finite audit family A, put $S_A = \mathrm{span}\{a_\ell : \ell \in A\}$. Equation S6 gives the exact worst-case separation

$$\Delta_A(k_N) = \mathrm{dist}_1(k_N, S_A). \tag{S46}$$

Because $S_A \subseteq \mathcal{P}_{m-1}$,

$$\Delta_A(k_N) \geq \mathrm{dist}_1(k_N, \mathcal{P}_{m-1}) = B_{m,N}. \tag{S47}$$

The complete prefix attains equality because $\mathrm{span}\{k_1, \ldots, k_m\} = \mathcal{P}_{m-1}$.

The same conclusion holds for an adaptive population-mean query rule that selects its next q_ℓ after observing earlier reliability means, while every query remains in Eq. S45. Under the endpoint worlds $f_\pm$, every such query returns

$$\langle a_\ell, f_+\rangle = \langle a_\ell, f_-\rangle = \sum_{n=1}^{m} q_{\ell n}\frac{1}{2} = \frac{1}{2}. \tag{S48}$$

The two worlds therefore generate the same population transcript under any such rule while remaining separated by $B_{m,N}$ at deployment. This proves the low-width inclusion statement in the main text. The restriction is essential. Score-dependent within-run stopping, finite-sample outcome-adaptive coefficient choice, an arbitrary selected-state kernel merely constrained to generate at most m candidates, direct width-N labels, candidate-level rank–truth observations, or another intervention kernel need not have the form in Eq. S45 and either needs separate protection or can enlarge the span. The mathematical step is span inclusion; the validation consequence is that knowing the complete low-width prefix is a favorable, information-maximal benchmark within this precisely stated mean-query class rather than a weak audit schedule. □

S5. Jacobi-theta phase transition and audit law

Set $\tau_{m,N} = m^2/N$ and assume $m, N \to \infty$ with $\tau_{m,N} \to \tau \in (0,\infty)$. For each fixed r,

$$\begin{aligned}\cos^{2N}\left\{\frac{r\pi}{2(m+1)}\right\} &= \exp\left[2N \log\cos\left\{\frac{r\pi}{2(m+1)}\right\}\right]\\ &\longrightarrow \exp\left(-\frac{\pi^2 r^2}{4\tau}\right).\end{aligned} \tag{S49}$$

To justify passage through the growing sum, define $a_{m,N,r} = \mathbf{1}\{r \leq m\}\cos^{2N}\{r\pi/[2(m+1)]\}$ for every $r \geq 1$. Because $N/(m+1)^2 \to 1/\tau$, for all sufficiently large m, $N/(m+1)^2 \geq 1/(2\tau)$. The elementary inequality $\cos x \leq e^{-x^2/2}$ on $[0,\pi/2]$ then gives, uniformly over every $r \geq 1$,

$$0 \leq a_{m,N,r} \leq \exp\left(-\frac{\pi^2 r^2}{8\tau}\right). \tag{S50}$$

The right side is summable on the positive integers, while Eq. S49 gives pointwise convergence for each fixed r. Dominated convergence on counting measure therefore yields

$$B_{m,N} \longrightarrow 1 + 2\sum_{r=1}^{\infty}(-1)^r e^{-\pi^2 r^2/(4\tau)} = \vartheta_4\left(0, e^{-\pi^2/(4\tau)}\right). \tag{S51}$$

Jacobi's imaginary transformation states

$$\vartheta_4(0, e^{-\pi t}) = t^{-1/2}\vartheta_2(0, e^{-\pi/t}). \tag{S52}$$

Taking $t = \pi/(4\tau)$ gives the positive, cancellation-free representation

$$B(\tau) = 4\sqrt{\frac{\tau}{\pi}}\sum_{j=0}^{\infty} e^{-\tau(2j+1)^2}. \tag{S53}$$

The first representation immediately gives, as $\tau \downarrow 0$,

$$B(\tau) = 1 - 2e^{-\pi^2/(4\tau)} + O(e^{-\pi^2/\tau}). \tag{S54}$$

The positive representation gives, as $\tau \to \infty$,

$$B(\tau) = 4\sqrt{\tau/\pi}\, e^{-\tau}\{1 + O(e^{-8\tau})\}. \tag{S55}$$

The inversion is now stated only for the limiting frontier. Define

$$\tau_\varepsilon = \inf\{\tau > 0 : B(\tau) \leq \varepsilon\}. \tag{S56}$$

Let $L = \log(1/\varepsilon)$. The large-τ expansion in Eq. S55 implies that inverting the leading relation

$$4\sqrt{\tau/\pi}\, e^{-\tau} = \varepsilon \tag{S57}$$

gives the asymptotics of τ_ε. Taking logarithms yields

$$\tau = L + \frac{1}{2}\log\tau + \log\frac{4}{\sqrt{\pi}}. \tag{S58}$$

Since $\tau/L \to 1$, one substitution yields

$$\tau = L + \frac{1}{2}\log L + \log\frac{4}{\sqrt{\pi}} + o(1), \tag{S59}$$

which proves the audit law in the main text. The order of limits is essential: for each fixed τ, first take $m, N \to \infty$ with $m^2/N \to \tau$ to obtain $B(\tau)$; only then let $\varepsilon \downarrow 0$ and invert that limiting frontier. The result does not assert that $B_{m,N} = B(m^2/N) + o(1)$ uniformly over a diverging phase coordinate, and hence does not cover an arbitrary joint sequence $\varepsilon = \varepsilon_N$ without an additional uniform remainder theorem. If the desired worst-case absolute error is δ rather than full interval width ε, use $\varepsilon = 2\delta$ within the same iterated regime.

For comparison, Eq. S44 satisfies

$$\log D_{m,N} = \sum_{j=1}^{m}\{\log(1-j/N) - \log(1+j/N)\} = -m(m+1)/N + o(1), \tag{S60}$$

so $D_{m,N} \to e^{-\tau}$. It gets the exponential order in the well-audited regime but not the exact prefactor or the under-audited frontier. For example, $B(1) \approx 0.8305$, whereas $e^{-1} \approx 0.3679$.

S6. Shape-constrained validation frontier

This section removes the unrestricted Chebyshev construction from the critical path. We derive the exact ambiguity for monotone reliability, an exact convex dual when monotonicity is combined with a Lipschitz bound, and matching-order lower and upper bounds. The techniques are classical measure representation, linear-programming duality, orthogonal polynomials, and Gauss quadrature; the resulting capped-tail validation dual and endpoint-resolution law are the contribution.

S6.1. Evidence-specific programs and the worst-evidence frontier

The main-text quantities $\Delta^{\mathcal{G}}_{m,N}$ are maxima over all feasible audit evidence. They are design guarantees, not the identified width for every realized audit vector. Put $b_n(t) = 1 - t^n$. For a feasible population audit vector $y = (y_1, \ldots, y_m)$, the exact monotone identified set is

$$\mathcal{I}_{\uparrow}(y) = \left\{ \int b_N\, d\alpha : \alpha \geq 0,\ \alpha([0,1]) \leq 1,\ \int b_n\, d\alpha = y_n\ (n \leq m) \right\}. \tag{S61}$$

For $0 < L < \infty$, the exact monotone-Lipschitz identified set is

$$\begin{aligned}\mathcal{I}_{\uparrow,L}(y) = \Big\{ c + \int_0^1 b_N(t)q(t)\,dt : c \geq 0,\ 0 \leq q \leq L,\ c + \int_0^1 q \leq 1,\\ c + \int_0^1 b_n(t)q(t)\,dt = y_n \quad (n \leq m) \Big\}.\end{aligned} \tag{S62}$$

Both are compact intervals whenever feasible; minimizing and maximizing the displayed target gives their endpoints. Their diameters are bounded above by the corresponding worst-evidence frontiers, with equality for at least one feasible y by compactness of the pair program. If audit means are known only within simultaneous confidence bands, replacing each equality by its band gives a valid finite-sample outer program. Sampling coverage is then a separate layer and is not supplied by the population frontiers themselves.

Lemma S3 (No duality gap for the equality-constrained measure programs)**.** *Take a compact Hausdorff space X and view the finite signed regular Borel measures as $\mathcal{M}(X) = C(X)^*$ with the weak-star topology. Let $\mathcal{K} \subset \mathcal{M}(X)$ be nonempty, convex, and weak-star compact, and let $b_0, b_1, \ldots, b_m \in C(X)$. Writing $\mathcal{C} = \mathcal{K} - \mathcal{K}$ and $h_{\mathcal{K}}(z) = \sup_{\mu\in\mathcal{K}} \int z\, d\mu$, one has*

$$\begin{aligned}&\sup_{\substack{\nu\in\mathcal{C}\\ \int b_j\, d\nu = 0,\ j=1,\ldots,m}} \int b_0\, d\nu\\ &= \inf_{a\in\mathbb{R}^m}\left\{ h_{\mathcal{K}}\left(b_0 - \sum_{j=1}^{m} a_j b_j\right) + h_{\mathcal{K}}\left(-b_0 + \sum_{j=1}^{m} a_j b_j\right)\right\}.\end{aligned} \tag{S63}$$

Because $\mathcal{C}$ is symmetric, the left side also equals the largest absolute separation subject to the equality constraints.

Proof. The difference set $\mathcal{C}$ is convex, symmetric, and weak-star compact. Both

$$A\nu = \left(\int b_1\, d\nu, \ldots, \int b_m\, d\nu\right), \qquad c(\nu) = \int b_0\, d\nu$$

are weak-star continuous. Consequently $G = \{(A\nu, c(\nu)) : \nu \in \mathcal{C}\} \subset \mathbb{R}^{m+1}$ is compact and convex, and $D = A\mathcal{C}$ is compact, convex, and symmetric. Symmetry implies $0 \in \operatorname{ri} D$, where relative interior is taken in $\operatorname{aff} D$; this also covers linearly dependent moment coordinates.

For $x \in D$, define

$$\phi(x) = \max\{c(\nu) : \nu \in \mathcal{C},\ A\nu = x\}.$$

The maximum exists because each fiber is compact. Compactness of G makes ϕ upper semicontinuous, and convexity of G makes ϕ concave. A finite concave upper-semicontinuous function has a supergradient at every relative-interior point of its domain. Thus there is an a on $\operatorname{aff} D$ (extended arbitrarily to $\mathbb{R}^m$) such that

$$\phi(x) \le \phi(0) + a^\top x, \qquad x \in D.$$

Writing $v = \phi(0)$, this inequality says $c(\nu) - a^\top A\nu \le v$ for every $\nu \in \mathcal{C}$. A maximizer in the fiber $A\nu = 0$ gives equality, so

$$\max_{\substack{\nu\in\mathcal{C}\\ A\nu=0}} c(\nu) = \inf_{a\in\mathbb{R}^m} \sup_{\nu\in\mathcal{C}}\{c(\nu) - a^\top A\nu\}. \tag{S64}$$

There is therefore no duality gap and the primal maximum is attained. Finally, $h_{\mathcal{K}-\mathcal{K}}(z) = h_{\mathcal{K}}(z) + h_{\mathcal{K}}(-z)$ gives Eq. S63; symmetry of $\mathcal{C}$ converts the one-sided maximum to the largest absolute separation. □

S6.2. Exact worst-evidence frontier under monotonicity

Let $\mathcal{F}_\uparrow$ be the nondecreasing measurable functions $f : [0,1] \to [0,1]$, with changes on null sets ignored, and define

$$\Delta^\uparrow_{m,N} = \sup_{\substack{f,g\in\mathcal{F}_\uparrow\\ \int k_n f = \int k_n g,\ n=1,\ldots,m}} \left|\int_0^1 k_N(u)\{f(u) - g(u)\}\, du\right|. \tag{S65}$$

Every right-continuous representative of $f \in \mathcal{F}_\uparrow$ is the distribution function of a subprobability measure α on $[0,1]$: $f(u) = \alpha([0,u])$. A jump at $u = 1$ changes the function only on a null set. Equivalently, its atom at $t = 1$ may be retained to keep the subprobability set weak-star compact: every audit and target integrand $1 - t^n$ vanishes there, so the atom changes neither constraints nor objective. Fubini's theorem gives

$$\theta_n(f) = \int_0^1 nu^{n-1} f(u)\, du = \int_{[0,1]} (1 - t^n)\, \alpha(dt). \tag{S66}$$

For coefficients $a = (a_1, \ldots, a_m)$, set

$$z_a(t) = 1 - t^N - \sum_{n=1}^m a_n(1 - t^n). \tag{S67}$$

Let $\mathcal{K}_\uparrow = \{\alpha \ge 0 : \alpha([0,1]) \le 1\}$, a weak-star compact convex set. The equality-constrained primal program is explicitly

$$\sup_{\substack{\alpha,\beta\in\mathcal{K}_\uparrow\\ \int(1-t^n)d(\alpha-\beta)=0,\ n\le m}} \int (1 - t^N) d(\alpha - \beta). \tag{S68}$$

For z_a in Eq. S67, the support of the unconstrained difference set in the residual direction is

$$\begin{aligned}&\sup_{\alpha([0,1])\le 1,\, \beta([0,1])\le 1} \int z_a\, d(\alpha - \beta)\\ &\quad = \max\{0, \sup_t z_a(t)\} + \max\{0, -\inf_t z_a(t)\} = \operatorname{osc}(z_a),\end{aligned} \tag{S69}$$

where the last equality uses $z_a(1) = 0$. Applying Lemma S3 to $b_0(t) = 1 - t^N$ and $b_n(t) = 1 - t^n$ gives, without a duality gap,

$$\Delta^\uparrow_{m,N} = \inf_{a\in\mathbb{R}^m} \operatorname{osc}(z_a). \tag{S70}$$

Every z_a is $p(t) - t^N$ for a polynomial $p \in \mathcal{P}_m$ satisfying $p(1) = 1$. Conversely, shifting any $p \in \mathcal{P}_m$ by the constant $1 - p(1)$ enforces that endpoint constraint without changing the oscillation of $p - t^N$. Finally, for every continuous function r,

$$\inf_{c\in\mathbb{R}} \|r - c\|_\infty = \frac{1}{2}\operatorname{osc}(r). \tag{S71}$$

Because $\mathcal{P}_m$ contains the constants, Eqs. S70 and S71 prove

$$\boxed{\Delta^{\uparrow}_{m,N} = 2E_m(u^N;[0,1]) := 2\inf_{p\in\mathcal{P}_m} \|u^N - p(u)\|_\infty.} \tag{S72}$$

Extremal monotone worlds may be taken as step functions. Here is the finite argument. For any nonzero feasible measure α with mass s, the vector of normalized integrals of $(b_1,\ldots,b_m,b_N)$ lies in the convex hull of the continuous curve $t\mapsto (b_1(t),\ldots,b_m(t),b_N(t))\subset\mathbb{R}^{m+1}$. Carathéodory's theorem represents that vector using at most $m+2$ curve points. Multiplying their weights by s preserves the mass, all m audits, and the deployment objective. The zero measure is already atomic. Applying this replacement separately to both measures in any optimal pair—whose existence follows from weak-star compactness—gives two optimal atomic measures, hence two right-continuous distribution functions with at most $m+2$ jumps each. Equation S72 is an exact validation consequence; best uniform approximation of monomials and its degree-squared resolution are classical [24, main result, pp. 451–455].

S6.3. Exact capped-tail dual under monotonicity and Lipschitz regularity

For $0<L<\infty$, let $\mathcal{F}_{\uparrow,L}$ contain the nondecreasing $[0,1]$-valued functions with Lipschitz constant at most L. Every member has the unique almost-everywhere representation

$$f(u) = c + \int_0^u q(t)\,dt, \qquad c\ge 0,\quad 0\le q\le L,\quad c+\int_0^1 q\le 1. \tag{S73}$$

For any integrable residual kernel r, write $Q_r(t)=\int_t^1 r(u)\,du$. Then

$$\int_0^1 r(u)f(u)\,du = cQ_r(0) + \int_0^1 Q_r(t)q(t)\,dt. \tag{S74}$$

Define the capped-tail support functional

$$H_L(z) = \sup_{\substack{c\ge0,\ 0\le q\le L\\ c+\int q\le 1}} \left\{ cz(0) + \int_0^1 z(t)q(t)\,dt \right\}. \tag{S75}$$

It has the explicit scalar-threshold representation

$$\boxed{H_L(z) = \inf_{\lambda\ge 0}\left\{\lambda + [z(0)-\lambda]_+ + L\int_0^1 [z(t)-\lambda]_+\,dt\right\}.} \tag{S76}$$

To prove it, introduce an item space $\{\star\}\cup[0,1]$ with capacity measure $\mu_L=\delta_\star + L\,dt$, item value $v(\star)=z(0)$ and $v(t)=z(t)$, and allocation $w(\star)=c$, $w(t)=q(t)/L$. Then $0\le w\le 1$ and $\int w\,d\mu_L\le 1$. For every $\lambda\ge 0$,

$$\int vw\,d\mu_L \le \lambda + [v(\star)-\lambda]_+ + L\int_0^1 [v(t)-\lambda]_+ dt.$$

Conversely, allocate full capacity to values strictly above a threshold, none below it, and just enough of the threshold level—including the atom when it ties—to fill at most one unit. If all positive capacity has mass at most one, the threshold is zero. This continuous-knapsack rule attains the infimum and proves Eq. S76. It also supplies an algorithm: for the polynomial residual z_a, find the roots of $z_a(t)=\lambda$, integrate its polynomial antiderivative on the positive intervals, and minimize one convex scalar function. The baseline atom competes through the separate $[z_a(0)-\lambda]_+$ term.

Bounded-density moment problems have a substantial classical theory, including necessary-and-sufficient compact-set conditions [20, Theorem 3(a–c), pp. 5–6]; here the functional arises as the exact dual norm of a validation residual.

For completeness, encode Eq. S73 by the measure

$$\nu = c\delta_0 + q(t)\,dt, \qquad c\ge 0,\quad 0\le q\le L,\quad \nu([0,1])\le 1. \tag{S77}$$

The atom at the left endpoint represents the baseline c and has coefficient $z(0)$; the absolutely continuous part represents the derivative. The set $\mathcal{K}_L$ of such measures is convex and weak-star compact: $c\in[0,1]$, the order interval $0\le q\le L$ is weak-star compact in L^∞, and the mass inequality is closed. Its support function is exactly $H_L(z)$. The equality-constrained program over two copies of $\mathcal{K}_L$ reproduces the audit agreement and deployment separation for two members of $\mathcal{F}_{\uparrow,L}$. Since $Q_{k_N-\sum a_n k_n}=z_a$, Lemma S3 therefore gives, without a duality gap,

$$\boxed{\Delta^{\uparrow,L}_{m,N} = \inf_{a\in\mathbb{R}^m}\{H_L(z_a)+H_L(-z_a)\}.} \tag{S78}$$

Unlike a qualitative assertion that smoothness helps, Eq. S78 is a computable finite-dimensional convex optimization over audit coefficients; evaluating H_L only requires thresholding the values of a one-dimensional tail residual. As L becomes unbounded, the derivative mass may concentrate arbitrarily and the sum of supports tends to $\operatorname{osc}(z_a)$, recovering Eq. S70.

S6.4. Explicit smooth indistinguishable worlds

Let $L_m(u) = P_m(2u-1)$ be shifted Legendre and set

$$A_{m,L} = \min\left\{\frac{1}{m+1}, \frac{L}{m(m+1)}\right\}. \tag{S79}$$

Define

$$\begin{aligned} f_m(u) &= A_{m,L}[L_m(0)]_+ + A_{m,L}\int_0^u [L'_m(t)]_+\,dt, \\ g_m(u) &= A_{m,L}[-L_m(0)]_+ + A_{m,L}\int_0^u [-L'_m(t)]_+\,dt. \end{aligned} \tag{S80}$$

Then $f_m - g_m = A_{m,L}L_m$. Both functions are nonnegative and nondecreasing. The shifted derivative satisfies

$$\|L'_m\|_\infty = m(m+1), \tag{S81}$$

so each Lipschitz constant is at most L. To check the upper range constraint, note that L_m has m monotonicity intervals and $|L_m| \le 1$, hence its total variation is at most $2m$. Splitting positive and negative variation and accounting for the initial value shows that each right side of Eq. S80 is at most $A_{m,L}(m+1) \le 1$. Thus $f_m, g_m \in \mathcal{F}_{\uparrow,L}$.

Orthogonality of L_m to $\mathcal{P}_{m-1}$ gives

$$\theta_n(f_m) - \theta_n(g_m) = A_{m,L}\int_0^1 k_n(u)L_m(u)\,du = 0, \qquad n \le m. \tag{S82}$$

At deployment, Eq. S44 gives the exact separation

$$|\theta_N(f_m) - \theta_N(g_m)| = A_{m,L}D_{m,N}, \qquad D_{m,N} = \prod_{j=1}^{m}\frac{N-j}{N+j}. \tag{S83}$$

This proves the lower bound using smooth monotone worlds rather than a sign oscillation.

S6.5. Positive-polynomial recovery and the upper bound

Let $f, g \in \mathcal{F}_{\uparrow,L}$ agree on the first m audits and put $h = f - g$. For $x < y$, both increments $f(y) - f(x)$ and $g(y) - g(x)$ lie in $[0, L(y-x)]$; their difference therefore lies in $[-L(y-x), L(y-x)]$. Thus h is L-Lipschitz, not merely $2L$-Lipschitz. Audit agreement implies

$$\int_0^1 u^j h(u)\,du = 0, \qquad j = 0, \dots, m-1. \tag{S84}$$

Put $d = \lfloor (m-1)/2 \rfloor$. Among nonzero $p \in \mathcal{P}_d$, consider the Rayleigh quotient

$$\rho_m = \min_{p \ne 0} \frac{\int_0^1 (1-u)p(u)^2\,du}{\int_0^1 p(u)^2\,du}. \tag{S85}$$

The eigenvalues of multiplication by u compressed to $\mathcal{P}_d$ are the shifted Gauss–Legendre nodes. Consequently

$$\rho_m = 1 - \xi_{d+1}, \tag{S86}$$

where ξ_{d+1} is the largest shifted Legendre zero of degree $d+1$. Let p attain Eq. S85 and normalize $q = p^2/\int p^2$. Then $q \ge 0$, $\int q = 1$, and $\deg q \le m-1$. Equation S84 gives $\int qh = 0$, and Lipschitz continuity gives

$$|h(1)| = \left|\int_0^1 q(u)\{h(1) - h(u)\}\,du\right| \le L\rho_m. \tag{S87}$$

If U_N has density k_N, then $\mathbb{E}(1 - U_N) = 1/(N+1)$. Hence

$$\begin{aligned} |\theta_N(f) - \theta_N(g)| &\le |h(1)| + \int_0^1 k_N(u)|h(u) - h(1)|\,du \\ &\le L\left(1 - \xi_{d+1} + \frac{1}{N+1}\right). \end{aligned} \tag{S88}$$

The unrestricted benchmark also gives $|\theta_N(f) - \theta_N(g)| \le B_{m,N}$, and bounded truth gives the trivial upper bound one. Combining Eqs. S83 and S88 proves the finite bounds in the shape-constrained corollary of the main text.

Finally, if $0 < L < \infty$ is fixed and $m^2/N \to \tau$, then $A_{m,L}m^2/L \to 1$ and Eq. S60 gives $D_{m,N} \to e^{-\tau}$. The classical extreme-zero asymptotic for Legendre polynomials gives

$$m^2(1 - \xi_{d+1}) \longrightarrow j_{0,1}^2, \tag{S89}$$

while $m^2/(N+1) \to \tau$ [25]. Multiplying the two bounds by m^2/L proves

$$e^{-\tau} \le \liminf \frac{m^2 \Delta_{m,N}^{\uparrow,L}}{L} \le \limsup \frac{m^2 \Delta_{m,N}^{\uparrow,L}}{L} \le j_{0,1}^2 + \tau. \tag{S90}$$

The two sides differ by constants depending only on the fixed phase coordinate, so the rate L/m^2 is sharp.

S6.6. Numerical evaluation and Lipschitz sensitivity

The reproduction script independently checks: (i) the first m Legendre moments by adaptive quadrature; (ii) the derivative bound $A_{m,L}m(m+1) \le L$; (iii) the range of both Jordan worlds on a 200,001-point grid; and (iv) the lower and Gauss-node upper bounds. Across 30 (m, N) cases with $L = 1$, the maximum audited-moment residual was 6.356×10^{-17}, the largest constructed Lipschitz constant was one, and the largest constructed world value was 0.417.

Table S3: Shape-restricted bounds at $m^2/N = 1$ and $L = 1$. Values are normalized by m^2/L; they tend toward the lower constant $e^{-1} = 0.3679$ and upper constant $j_{0,1}^2 + 1 = 6.7832$.

m	$m^2 A_{m,1} D_{m,N}$	$m^2\{1 - \xi_{d+1} + 1/(N+1)\}$
4	0.225	4.322
8	0.288	5.428
16	0.325	6.079
32	0.346	6.426
64	0.357	6.603

To evaluate the exact capped-tail dual itself, we restricted each derivative q to be constant on nested equal-width percentile bins. This gives a finite primal LP over two admissible piecewise-linear reliability laws and a bin-averaged version of Eqs. S78 and S76. In exact arithmetic the two finite LP formulations are duals. The returned floating-point coefficient vector a was then evaluated against the *continuum* H_L by root-finding for $z_a(t) = \lambda$ and analytic integration of the polynomial residual between roots. Exact feasibility would make the bin construction a lower bound, and exact evaluation at any a would be an upper bound. The stored solver vectors, roots, and integrals were not wrapped in interval arithmetic, however, so the following pairs are numerical estimates rather than certified enclosures.

Table S4: Numerical primal–dual estimates for the exact capped-tail frontier. The primal uses 8,192 bins; "dual" evaluates the continuum threshold functional at the returned audit coefficients. "Gap" is their floating-point difference, not an interval-certified enclosure width. The final column is the rigorous but looser analytic bracket in the main corollary.

(m, N, L)	Primal	Dual	Gap	Constructive bracket
$(2, 8, 1)$	0.139322917	0.139322920	3.49×10^{-9}	[0.07778, 0.61111]
$(4, 16, 1)$	0.046130661	0.046130667	6.41×10^{-9}	[0.01409, 0.27015]
$(8, 64, 1)$	0.020791486	0.020791501	1.49×10^{-8}	[0.00449, 0.08482]
$(8, 64, 4)$	0.072274153	0.072274347	1.94×10^{-7}	[0.01798, 0.33927]

Across these rows the numerical finite-LP primal–dual gap is at most 2.1×10^{-12}. Doubling from 4,096 to 8,192 bins changes the primal by at most 2.84×10^{-7}. For $(m, N, L) = (4, 16, 1)$ one returned optimizer is

$$a = (-0.931448,\ 7.151953,\ -15.935844,\ 10.715339),$$

with both thresholds zero. For $(8, 64, 4)$ the positive-residual threshold is 0.0228384 and the negative threshold is zero, illustrating the competition between derivative capacity and the baseline atom. Power-basis audit coefficients become ill-conditioned at $m = 8$ even though the objective is stable; all coefficients and nested-grid values are supplied in `results/capped_tail_dual_coefficients.csv` and `results/capped_tail_dual_checks.csv`. The independent script is `code/solve_capped_tail_dual.py`.

The percentile coordinate gives L a direct scale interpretation: $|f(u+h) - f(u)| \le Lh$. A reliability change of one therefore requires at least $1/L$ of the rank range; for $L < 1$ it cannot occur anywhere on $[0, 1]$. The audited moment prefix does not validate this local assumption. It should be bounded using scientific knowledge or independent candidate-level rank–truth data with uniform uncertainty, and otherwise varied as a sensitivity parameter.

Table S5: Sensitivity to the percentile-scale Lipschitz bound at $(m, N) = (8, 64)$. Values are converged floating-point primal–dual estimates, not interval-certified enclosures; the last row is the separate monotone frontier $2E_8(u^{64})$.

L	Minimum rank span $1/L$	$\Delta^{\uparrow,L}_{8,64}$
0.25	4.0000	0.005198
0.50	2.0000	0.010396
1	1.0000	0.020791
2	0.5000	0.041509
4	0.2500	0.072274
8	0.1250	0.108038
16	0.0625	0.138346
∞	0	0.162160

The nearly linear regime through $L = 1$ occurs because the total-mass constraint in H_L is not yet active. At larger L, the optimal threshold becomes positive and the frontier bends toward the monotone limit. These values show why reporting only the analytic brackets can be misleading and why conclusions should be accompanied by an L sensitivity curve.

S7. Extensions and scope

S7.1. Bounded and categorical outcomes

Nothing in the inverse theorem requires binary truth. For $T \in [a, b]$, rescale to $[0, 1]$ and set $f(x) = \mathbb{E}(T \mid X = x)$. All ambiguity results then rescale by $b - a$. For categories $Y \in \{1, \ldots, C\}$, define $f_c(x) = \mathbb{P}(Y = c \mid X = x)$. The geometry applies coordinatewise subject to $\sum_c f_c = 1$; joint identified sets may exploit that simplex constraint.

S7.2. Known-kernel selection laws beyond iid search

Best-of-N is one kernel family. A recent test-time-scaling taxonomy distinguishes single-trajectory sequential scaling, leaf-level terminal reduction, and prefix-level search [26]. The closed form in this paper covers the fixed-proposal iid special case of leaf-level maximum-score reduction. A randomized shortlist, top-k draw, rejection sampler, beam-search trace, or dependent candidate process induces another selected-state law k_w. If that law is known or can be estimated independently on the same stable state space, Eq. S6 applies without an iid assumption and returns $\mathrm{dist}_1(k_w, S_V)$; it does not return $B_{m,N}$. Dependence among candidates changes the selected-rank kernel and may change effective search breadth in either direction. Adaptive generators create history-dependent kernels; the natural state x must then include the relevant history, and validation must span the conditional deployment functionals. In particular, one must not substitute $k_N(u) = Nu^{N-1}$ for an agent that updates its generator after observing intermediate scores. If the policy-induced law is unknown or candidate generation changes the reliability surface, neither the exact best-of-N frontier nor the general known-kernel identity supplies a certificate. The generality is therefore exact but conditional: it covers known kernels, not unspecified adaptive agents.

The boundary need not be binary when the policy law is estimable. Let $v_1, \ldots, v_p, k$ be modeled normalized kernels and let $\widetilde{v}_1, \ldots, \widetilde{v}_p, \widetilde{k}$ be the actual normalized kernels on the same surface, with

$$\|\widetilde{v}_j - v_j\|_1 \le \delta_j, \qquad \|\widetilde{k} - k\|_1 \le \delta_k. \tag{S91}$$

Proposition S1 (Kernel-perturbation certificate)**.** *If $f, g \in \mathcal{F}$ agree under every actual audit kernel, then*

$$|\langle \widetilde{k}, f - g\rangle| \le \inf_{a \in \mathbb{R}^p} \left\{ \left\| k - \sum_{j=1}^p a_j v_j \right\|_1 + \delta_k + \sum_{j=1}^p |a_j| \delta_j \right\}. \tag{S92}$$

For one world, the centered reconstruction from the actual audit means $\widetilde{y}_j = \langle \widetilde{v}_j, f\rangle$ satisfies, for every a,

$$\left| \langle \widetilde{k}, f\rangle - \left\{ \frac{1}{2} + \sum_{j=1}^p a_j (\widetilde{y}_j - \tfrac{1}{2}) \right\} \right| \le \frac{1}{2} \left\{ \left\| k - \sum_{j=1}^p a_j v_j \right\|_1 + \delta_k + \sum_{j=1}^p |a_j| \delta_j \right\}. \tag{S93}$$

Proof. For $h = f - g$, actual audit agreement gives $\langle \widetilde{v}_j, h\rangle = 0$ and $\|h\|_\infty \le 1$. For any a,

$$|\langle \widetilde{k}, h\rangle| \le \|\widetilde{k} - k\|_1 + \left\| k - \sum_j a_j v_j \right\|_1 + \sum_j |a_j| \|v_j - \widetilde{v}_j\|_1, \tag{S94}$$

which proves Eq. S92 after taking the infimum. For Eq. S93, write both the actual target and audit means against $f - 1/2$, whose L^∞ norm is at most $1/2$, and apply the same triangle inequality. $\square$

Deterministic audit-mean errors $|\widehat{y}_j - \widetilde{y}_j| \le \eta_j$ add $\sum_j |a_j|\eta_j$ to the right side of Eq. S93; the finite-task stochastic term in Eq. S27 is then added separately. Robust optimal recovery from inaccurate data is established more generally [27]; no new abstract perturbation theorem is claimed. The validation-specific use is to turn an independently estimated policy law into an auditable penalty. If a logged adaptive policy yields confidence bounds for its history-distribution kernel, Eq. S93 quantifies the remaining model risk. Without such bounds, describing the generator as adaptive supplies no certificate.

S7.3. Structural assumptions can narrow the frontier

The Chebyshev worlds oscillate because the model class permits every measurable $f \in [0, 1]$. Monotonicity, Lipschitz bounds, bounded variation, analyticity, or a parametric tail law can sharply reduce ambiguity. Such assumptions should be represented by replacing $\mathcal{F}$ in Eq. S1 and recomputing its support function. They are scientifically meaningful only if justified and stress-tested prospectively. Replication at the same interventions cannot manufacture them.

S7.4. Unknown kernels and distribution shift

The span identity treats v_j and k_w as known and assumes that they act on a common reliability surface. Proposition S1 handles bounded kernel error on that surface; sample splitting or joint empirical-process bounds can supply the δ_j. If deployment changes the underlying reliability surface from f to f_w, however, kernel geometry alone is insufficient. One then needs a transport or invariance model relating the surfaces. The present result should be read as a lower-level identification limit, not a distribution-shift theorem: failure already occurs with known kernels and a common, stable f.

S8. Mathematical-reasoning search experiment

Empirical purpose and external-validity boundary. The experiments are retrospective, contrastive demonstrations of the modeled mechanism, not estimates of its prevalence across AI systems and not empirical validation of the distribution-free extremizers. Within each model–task cell, the candidates are a fixed finite empirical population. The iid idealization is conditional on task: task t has its own randomized percentile and law f_t, so $\theta_{n,t} = \int k_n f_t$. Equal-weight macro averaging gives $\overline{\theta}_n = \int k_n \overline{f}$ with $\overline{f} = T^{-1}\sum_t f_t$; no exchangeability across tasks is assumed. Every reported best-of-n estimand is the exact with-replacement maximum-score functional of the corresponding task-specific population with randomized ties; it is not an observed maximum among n distinct newly generated candidates. The analyses test four narrower claims: changing that score-and-select functional changes the truth target; task-level reversals can coexist with favorable mean scaling; a finite empirical audit prefix can leave a wide compatible deployment interval; and score-matched label acquisition can reduce finite estimation error relative to uniform labels within one fixed pool. The GSM8K analysis uses natural-language answers, consensus ranking, exact-answer truth, and widths through 4,096. The independent CodeRM analysis in Section S10 uses programs, generated-unit-test ranking, separate HumanEval+ truth, and widths through 100. This variation supports construct-level replication within the conditional-iid score-and-select setting. It does not make either task collection representative of adaptive agents, new-candidate deployment, subjective evaluation, deployment frequency, or the probability of a near-worst-case world.

S8.1. Dataset and pinning

The public Monkey Business dataset contains 10,000 generated GSM8K solutions per problem and model [21]. We used 127 problems for each of two configurations:

- `GSM8K_Llama-3-8B-Instruct.json`; SHA-256 `0e334c7010a2eb39ab0aaf38cdd196da0b6219a95fd69f82d35b8f51e46ed765`;
- `GSM8K_Llama-3-70B-Instruct.json`; SHA-256 `22ff8a363d5a5b8ea5a6c299285d8f615ec32b9c1fe08e8253cc1a6c9c7c323f`.

Both files are downloaded from revision `a9f8f73bcd6948a57ed922cba4e48062ef95f553`. The pinned release file, rather than an outcome-dependent rule applied by us, fixes the inclusion set: each file contains exactly 127 problem records, and we included every record in both files without excluding any eligible problem. Each record contains 10,000 generated texts and exact final-answer correctness indicators. Total analyzed candidates were $127 \times 10{,}000 \times 2 = 2.54$ million.

S8.2. Consensus score and disjoint deployment population

Within every model–problem cell, a deterministic random permutation used 2,000 solutions as the reference set and 8,000 as the deployment population. The split seed was 20,260,817 plus fixed model and problem offsets. The final answer following the GSM8K delimiter was normalized by removing commas and whitespace. Its frequency in the reference set was the consensus score for deployment candidates with that answer; unparsable answers received a score below every parsed answer.

For a discrete score group ℓ, let $F_{\ell-}$ and F_ℓ be its lower and upper cumulative population masses and let t_ℓ be its mean truth label. Under iid sampling with replacement from the empirical deployment population and uniform tie breaking, the exact selected truth at width n is

$$\widehat{\theta}_n = \sum_\ell t_\ell \{F_\ell^n - F_{\ell-}^n\}. \tag{S95}$$

This avoids Monte Carlo simulation of search. We evaluated $n \in \{1, 2, 4, \ldots, 4096\}$.

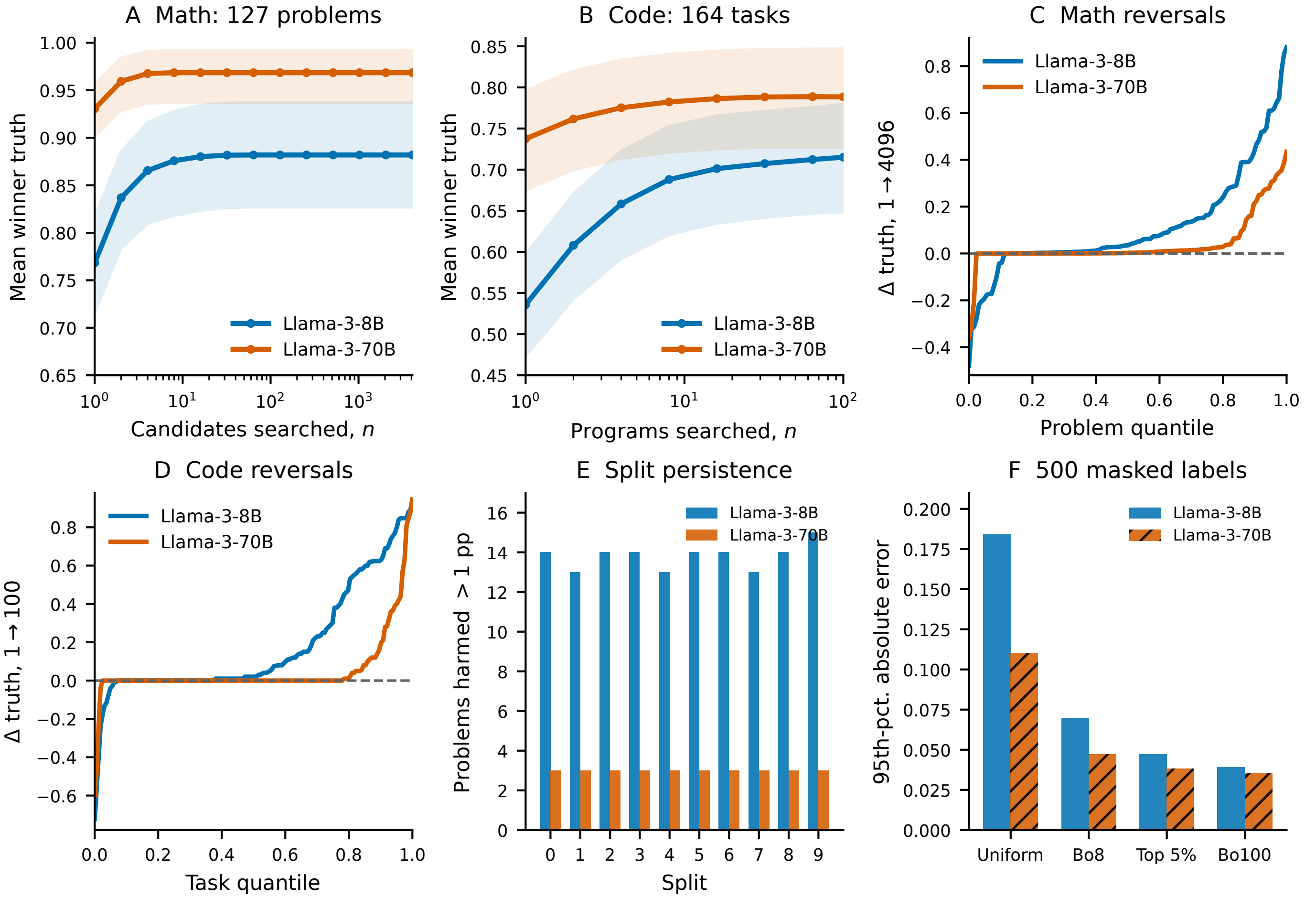

Figure S1: Retrospective construct checks in two fixed public candidate populations. (A) Mean selected truth and 95% problem-bootstrap intervals across 127 GSM8K problems under answer-consensus ranking. (B) The corresponding functional across 164 HumanEval+ tasks under CodeRM ranking. (C,D) Task-level changes include reversals despite favorable aggregate scaling. (E) GSM8K harm counts persist across 10 reference/deployment splits. (F) In masked-label replays at a 500-label budget, a score-tail rule frozen after an 82-task design split reduces held-out best-of-100 estimation error relative to uniform candidate labels. All best-of-n quantities are with-replacement functionals of fixed empirical populations. The panels are not prospective tests of audit effectiveness or prevalence estimates.

S8.3. Uncertainty and robustness

All reported aggregates treat the problem as the sampling unit. Percentile intervals use 20,000 bootstrap resamples of the 127 problem rows, with fixed seed 73,041 plus model offset. Macro AUC is the mean within-problem ROC AUC over deployment populations containing both truth classes: 125 problems for Llama-3-8B and 107 for Llama-3-70B. The remaining two and 20 single-class populations remain in all search-reliability and harm summaries but do not have a defined AUC. Ten robustness splits use distinct deterministic seeds and repeat reference scoring, deployment construction, exact winner integration, and harm counting. This checks that reversals are not artifacts of a particular 2,000/8,000 partition; it does not make the 127 GSM8K problems representative of every task domain.

These data validate three premises, not the distribution-free worst case itself: the selection kernel changes materially with search breadth; high average rank discrimination can coexist with problem-level reversals; and those reversals persist under resplitting. The Chebyshev theorem is a mathematical identification result and does not require empirical frequency of its extremizers.

S9. Finite and asymptotic efficiency of validation interventions

S9.1. Winner-distribution importance sampling

The target is

$$\theta_N = \int_0^1 k_N(u) f(u) du, \qquad k_N(u) = N u^{N-1}. \tag{S96}$$

Table S6: Empirical search-scale summary. Intervals are 95% problem-bootstrap percentile intervals. Harm counts compare $n = 4096$ with $n = 1$.

Quantity	Llama-3-8B	Llama-3-70B
Problems	127	127
Candidates	1,270,000	1,270,000
$\widehat{\theta}_1$	0.7684	0.9305
$\widehat{\theta}_{4096}$	0.8819	0.9685
Mean change	0.1135	0.0380
Change interval	[0.0748, 0.1538]	[0.0195, 0.0576]
Problems entering macro AUC	125	107
Macro AUC	0.9070	0.9617
AUC interval	[0.8616, 0.9474]	[0.9230, 0.9901]
Problems harmed > 1 pp	14	3
Problems harmed > 5 pp	12	3
Worst problem change	-0.4796	-0.3598
Best problem change	0.8803	0.4339
Harm > 1 pp over 10 splits	13–15	3 in every split

Suppose an audit draws a labeled winner from width s, whose percentile density is $q_s(u) = su^{s-1}$. The Horvitz–Thompson variable

$$Z_s = \frac{k_N(U)}{q_s(U)}T, \qquad U \sim q_s, \quad T \mid U \sim \text{Bernoulli}\{f(U)\}, \tag{S97}$$

is unbiased because

$$\mathbb{E}Z_s = \int q_s \frac{k_N}{q_s} f = \theta_N. \tag{S98}$$

Its second moment is

$$\begin{aligned} \mathbb{E}Z_s^2 &= \int_0^1 q_s(u) \left\{ \frac{k_N(u)}{q_s(u)} \right\}^2 f(u)du \\ &= \frac{N^2}{s} \int_0^1 u^{2N-s-1} f(u)du \\ &= \frac{N^2}{s(2N-s)} \theta_{2N-s}, \end{aligned} \tag{S99}$$

so

$$\text{Var}(Z_s) = \frac{N^2}{s(2N-s)} \theta_{2N-s} - \theta_N^2. \tag{S100}$$

For $s = N$, the weight is one and the variance is $\theta_N(1-\theta_N)$. We define the variance-equivalent label ratio

$$\rho_s = \frac{\text{Var}(Z_s)}{\theta_N(1-\theta_N)}. \tag{S101}$$

Under independent sampling and the same estimator class, ρ_s labels from design s have the same asymptotic variance as one deployment-winner label.

S9.2. Top-tail design

An audit uniform on the top fraction a of percentiles has density $q_a(u) = a^{-1}\mathbf{1}\{u \geq 1-a\}$. It targets the truncated functional

$$\theta_{N,a} = \int_{1-a}^1 k_N(u) f(u) du. \tag{S102}$$

The omitted nonnegative mass satisfies

$$0 \leq \theta_N - \theta_{N,a} \leq \int_0^{1-a} k_N(u)du = (1-a)^N. \tag{S103}$$

For $a = 0.05$ and $N = 4096$, this is $0.95^{4096} = 5.70 \times 10^{-92}$. Its importance-weighted second moment is

$$\mathbb{E}Z_a^2 = \frac{aN^2}{2N-1} \theta_{2N-1}^{[1-a,1]}, \tag{S104}$$

where the superscript denotes the contribution of the top interval to the corresponding selected-mean integral.

S9.3. Retrospective masked-label acquisition replay

The variance calculation above predicts a design advantage but does not show its finite-sample size. We therefore used the candidate-level CodeRM data in Section S10 as a label oracle. For every model–task cell, all truth values were treated as masked at the allocation stage. Each acquisition draw first sampled one of the 164 tasks uniformly, then selected a candidate using only its verifier score under one of four fixed rules: uniform candidate, best-of-8 winner, randomized top-5% score percentile, or best-of-100 deployment winner. Full truth was used only after acquisition to score the estimator against the exact macro best-of-100 target.

For a tied score group occupying sorted candidate indices $a+1,\ldots,b$ among $M = 100$ candidates, the probability assigned to each candidate under best-of-s selection is

$$q_s(i) = \frac{(b/M)^s - (a/M)^s}{b-a}. \tag{S105}$$

This is exact for with-replacement search, maximum-score selection, and uniform randomization within the winning tie. Let $p_i = q_{100}(i)$ denote the target probability and let q_i denote the audit probability within the sampled task. The revealed-label variable

$$Z_i = \frac{p_i}{q_i} T_i, \tag{S106}$$

is unbiased for the task-averaged best-of-100 target whenever $q_i > 0$ wherever $p_i > 0$. For the top-tail design, Eq. S106 estimates the truncated target; the missing mass is at most $0.95^{100} = 0.0059205$. Randomized percentile sampling distributes a boundary-crossing tie uniformly across the entire score group.

We ran 5,000 independent acquisition replicates for label budgets $25, 50, 100, 250, 500, 1{,}000, 2{,}000$, and $5{,}000$, using seed 131,071 plus fixed model and design offsets. Within a replicate, samples accumulate across budgets. The primary finite error metric is the 95th percentile across replicates of $|\widehat{\theta}_{100} - \theta_{100}|$; root mean squared and median absolute errors are also written to `results/coderm_label_acquisition.csv`.

Table S7: Finite masked-label error for best-of-100 CodeRM truth. Entries are the 95th percentile of absolute error across 5,000 retrospective replays. Allocation uses scores only; full labels score the emulation afterward.

Model	Audit design	100	500	1,000	5,000 labels
8B	Uniform candidate	0.444	0.184	0.131	0.060
8B	Best-of-8 winner	0.157	0.070	0.051	0.022
8B	Top-5% percentile	0.105	0.047	0.034	0.015
8B	Best-of-100 winner	0.085	0.039	0.028	0.013
70B	Uniform candidate	0.212	0.110	0.079	0.035
70B	Best-of-8 winner	0.109	0.047	0.033	0.015
70B	Top-5% percentile	0.085	0.038	0.027	0.012
70B	Best-of-100 winner	0.082	0.036	0.025	0.011

At 500 labels, replacing uniform acquisition by the top-tail design reduced the 95th-percentile error by factors 3.89 and 2.89 for 8B and 70B. The same top-5% rule, budgets, and estimator were used for both models; no model-specific truth information changes allocation. The actual top-tail truncation biases under the fully labeled empirical populations were 9.75×10^{-5} and 3.64×10^{-5}, far below the distribution-free bound, but those empirical values are used only for retrospective scoring. This is a finite label-acquisition emulation, not a prospective experiment with newly generated candidates or newly collected truths. A genuinely prospective test would randomize the audit design before outcomes are produced and should additionally account for per-design generation cost.

The replay errors and the 1,000-bin feasible spans in Section S10.2 are not one metric. The prefix programs construct compatible continuum step-function witnesses under exact or banded mean constraints; the replay asks how a fixed estimator behaves when labels are acquired in different score directions. Their joint conclusion is narrower than a prospective effectiveness claim: on the same best-of-100 functional, covered search widths, independent task count, and label direction address different failure modes and must be planned separately.

S9.4. Frozen-rule held-out audit test

The full-pool replay above fixes the allocation rule in advance but evaluates it on the same task population used to report the comparison. To test whether an audit rule selected from data transfers across tasks, we added a label-blind discovery/held-out split before examining outcomes. With seed 260,821, the 164 task identifiers shared by both CodeRM models were permuted once and divided into 82 discovery and 82 held-out tasks. The split is shared across models and is recorded in `results/coderm_heldout_task_split.csv`.

The candidate audit family was fixed at the top 5%, 10%, or 20% of score percentiles. All three designs leave at most $0.95^{100} = 0.0059205$ of the best-of-100 target mass unsupported. On discovery tasks only, we ran 5,000 masked-label replays at a budget of 500 and selected the fraction minimizing the mean 95th-percentile absolute error across the two models. Mean discovery errors were 0.0427, 0.0546, and 0.0751, respectively, so the 5% rule was selected. That rule was then frozen before the held-out target and replay error were computed. Uniform acquisition served as the held-out comparator. The estimator and exact tied-score candidate probabilities are those in Eqs. S105 and S106; independent replay streams use seed 262,147 plus fixed design and model offsets.

Table S8: Frozen-rule performance on 82 held-out CodeRM tasks. Entries are based on 5,000 retrospective masked-label replays at 500 labels. The top-tail fraction was chosen on a disjoint 82-task discovery split and frozen before held-out outcomes were scored.

Model	Held-out target	Uniform q95 error	Frozen top-5% q95 error	Frozen absolute bias
8B	0.698	0.180	0.049	7.58×10^{-5}
70B	0.791	0.076	0.035	7.28×10^{-5}

The frozen rule reduced the held-out q95 error by factors 3.67 and 2.15. This is stronger than selecting and evaluating a design on the same task set: held-out outcomes did not choose the tail fraction. It is nevertheless not a prospective intervention. Both splits come from one existing public dataset, all candidates and labels had already been generated, and the replay does not measure collection cost, distribution shift, or a causal effect of assigning a real evaluation program. Complete discovery and held-out metrics, including median error and RMSE, are in `results/coderm_heldout_audit.csv`.

S9.5. Asymptotic GSM8K calculation

Within each model, tie-averaged truth was aggregated into 20 equal-width percentile bins across problems. Treating the bin means as a piecewise-constant f, all moments in Eqs. S100 and S104 are analytic differences of bin-edge powers. This binning is deliberately coarse; at $N = 4096$, essentially all target mass lies in the last bin, and the computation is a transparent design comparison rather than a claim of nonparametric tail recovery.

Table S9: Variance-equivalent labels for a best-of-4096 target. Each entry is the number of labels under the row design needed to match the asymptotic variance of one deployment-winner label.

Audit design	Llama-3-8B	Llama-3-70B
Uniform candidate	17,333.7	64,993.0
Best-of-8 winner	2,162.0	8,104.2
Best-of-64 winner	265.6	993.1
Uniform top 5% rank	859.6	3,220.4
Best-of-4096 winner	1.0	1.0

The large ratios have a simple source. Under uniform candidate sampling, the importance weight near $u = 1$ is approximately N, so rare tail labels receive enormous leverage. The deployment-winner design samples directly from that tail and uses unit weights. The ratios depend on f, N, the score law, and the estimator; they are not transferable constants. In a real evaluation, cost per label, dependence among generated candidates, and the feasibility of producing deployment winners must also be included.

S9.6. Prospective intervention specification

The public-data experiment can be converted into a genuinely prospective test without changing its estimand. Before generating or labeling new outcomes, register the target width N, task population, verifier, tie rule, label budget, top-tail fraction, and the uniform-versus-score-matched allocation comparison. Randomize independent task–model cells to audit designs before truth is observed; retain the task as the cluster; and collect a fully labeled evaluation subset only to score estimator error, not to choose the allocation. The primary comparison should be finite-sample squared or absolute error at a fixed total cost, with coverage and generation cost as secondary outcomes. If the objective is instead a distribution-free direct-winner interval of full width 0.10 at 95% confidence, Eq. S28 requires 738 independent tasks; this is a conservative certificate calculation, not a power analysis. The fixed replay design used here—$N = 100$, top 5%, and the eight reported budgets—is a ready specification, but this paragraph is a protocol, not additional prospective evidence.

S10. Code-generation replication and reproducibility

S10.1. CodeRM/HumanEval+ data and estimand

The independent domain is the public CodeRM release [22], pinned at Git commit `aa4946e9245ed41e24d60ad29e965132b5b84fe6`. We used all 164 HumanEval+ tasks and the first 100 candidate programs per task from each of the Llama-3-8B and Llama-3-70B annotated solution files. The annotation SHA-256 digests are, respectively,

- `cae513c7578a52c6d3110a13bbf2514e44a0cd58f2a4a258b58e7337c2af0211`;

- 6ddb4273e4d9a35f7bef6016f37ac2e9b99d6bd7aad05357f7ff9399b0d45850.

Objective truth is the HumanEval+ `plus_status` field, which evaluates the program against the benchmark's expanded hidden tests.

The CodeRM output archive linked by the repository README has SHA-256 728780d642f465e91f8619de1bae0cb3 192519278e87014d546cee9921d4e6c5. For each model we used the public 100-solution by 100-generated-unit-test execution matrix. The extracted JSONL member digests are

- 8B: 8169be74c8144e0bad6b593093a02e4963144f66a6a5f6b84c10ff1dd6e316d1;
- 70B: 7edb5fc98033f5de748c5dfe2f740cdeffb02020ead8f49988ea822d2d8d2fdf.

For candidate i, the verifier score is the fraction of the 100 CodeRM-8B-generated unit tests for which the execution result is `pass`. The data therefore contain $164 \times 100 \times 2 = 32{,}800$ programs and 3.28 million candidate–test executions. The generated tests determine ranking; the distinct HumanEval+ status determines truth.

Within each model–task cell, Eq. S95 was applied to the empirical verifier-score groups, with exact uniform tie breaking, for $n \in \{1, 2, 4, 8, 16, 32, 64, 100\}$. Intervals use 20,000 task bootstrap resamples, with seed 92,771 plus model offset. AUC gives half credit to tied positive–negative pairs. All 164 tasks contain both truth classes for each model and therefore enter both macro AUCs. The task is the independent unit; the 100 generated tests do not count as 100 independent benchmark tasks.

Table S10: Independent CodeRM/HumanEval+ replication. Intervals are 95% task-bootstrap percentile intervals. Harm compares best-of-100 with one random program.

Quantity	Llama-3-8B	Llama-3-70B
Tasks	164	164
Candidate programs	16,400	16,400
Candidate–test executions	1,640,000	1,640,000
Tasks entering macro AUC	164	164
Macro AUC	0.931 [0.898, 0.959]	0.881 [0.818, 0.938]
$\widehat{\theta}_1$	0.536 [0.473, 0.597]	0.737 [0.675, 0.797]
$\widehat{\theta}_{100}$	0.715 [0.649, 0.780]	0.788 [0.726, 0.847]
Mean change	0.179 [0.134, 0.226]	0.051 [0.025, 0.080]
Tasks harmed by > 1 pp	10	4
Worst task change	−0.722	−0.600

The CodeRM replication is not a second estimate of the GSM8K mechanism. It deliberately changes the domain, generator outputs, score construction, truth mechanism, and search range. Its role is to test the more basic prediction that a favorable average verifier can induce a heterogeneous reliability spectrum with severe local reversals. The repository did not expose an explicit license file at the pinned revision, so the package does not redistribute programs, generated unit tests, or execution logs. It includes only derived numeric candidate scores and labels, plus a checksum-verifying reconstruction script.

S10.2. Retrospective compatibility from an audited prefix

The main-text certification exercise uses the same 32,800 derived CodeRM score–truth rows. Within task t, sort the 100 candidates by verifier score. If a score-tie group occupies percentile intervals I_{tj} and has mean truth $\bar{T}_{tj}$, define the bounded step function

$$\widehat{f}_t(u) = \bar{T}_{tj}, \qquad u \in I_{tj}. \tag{S107}$$

Spreading a tie group's mean over its occupied intervals is exactly equivalent to uniform tie breaking. Consequently the macro law $\widehat{f} = 164^{-1} \sum_t \widehat{f}_t$ satisfies

$$\int_0^1 n u^{n-1} \widehat{f}(u)\, du = \frac{1}{164} \sum_{t=1}^{164} \widehat{\theta}_{t,n} \tag{S108}$$

for every integer n, not only for the eight search widths previously plotted.

To ask what the audited prefix alone certifies, refine the 100 empirical rank intervals to $J = 1{,}000$ equal bins and let $g_j \in [0, 1]$ be a candidate law on bin j. Put

$$w_{n,j} = \left(\frac{j}{J}\right)^n - \left(\frac{j-1}{J}\right)^n. \tag{S109}$$

For each m, the plug-in endpoints solve the two linear programs

$$\underset{0 \le g \le 1}{\text{minimize/maximize}} \quad w_N^\top g \quad \text{subject to} \quad w_n^\top g = \widehat{\theta}_n, \quad n = 1, \ldots, m. \tag{S110}$$

Every optimizer defines a bounded piecewise-constant function on $[0, 1]$. Thus the displayed separation is a constructive lower bound on the unrestricted continuum identified width; it is not created by allowing infeasible grid vectors. Raw power-moment rows become nearly collinear as m grows. For exact plug-in equalities, the implementation therefore normalizes the rows and replaces them by an orthonormal basis obtained from a QR factorization of the transposed audit matrix. This changes neither the row space nor the exact feasible set, but prevents a tiny raw-moment tolerance from becoming a large extrapolated target error. At $m = 8$, increasing J from 200 to 500 and 1,000 changed every endpoint by less than 0.007. The maximum raw equality residual at $J = 1{,}000$ was 1.3×10^{-14}.

Finite task sampling uncertainty was propagated separately. We drew 20,000 multinomial bootstrap resamples of the 164 tasks. For each m, let $\widehat{s}_n$ be the bootstrap standard error and let c_m be the 95th percentile of

$$\max_{1\le n\le m} \left| \frac{\widehat{\theta}_n^* - \widehat{\theta}_n}{\widehat{s}_n} \right|. \tag{S111}$$

Replacing the equalities in Eq. S110 by the simultaneous inequalities $\widehat{\theta}_n - c_m\widehat{s}_n \le w_n^\top g \le \widehat{\theta}_n + c_m\widehat{s}_n$, truncated to $[0, 1]$, gives the bootstrap-feasible ranges in Table S11. They condition on the empirical score/rank construction and use the task as the sampling unit; they are not a distribution-shift guarantee.

Table S11: CodeRM best-of-100 ranges from audits through m. Plug-in columns impose exact empirical means at every integer $n \le m$. Bootstrap columns allow the simultaneous 95% task-bootstrap band. Observed best-of-100 truth is 0.715 for 8B and 0.788 for 70B.

Model	m	Plug-in range	Width	Bootstrap-feasible range
Llama-3-8B	4	[0.003, 1.000]	0.997	[0.000, 1.000]
	8	[0.186, 0.992]	0.806	[0.003, 1.000]
	12	[0.453, 0.907]	0.453	[0.021, 1.000]
	16	[0.620, 0.796]	0.176	[0.056, 1.000]
	20	[0.690, 0.737]	0.047	[0.102, 0.999]
Llama-3-70B	4	[0.018, 1.000]	0.982	[0.000, 1.000]
	8	[0.305, 0.996]	0.691	[0.016, 1.000]
	12	[0.569, 0.937]	0.367	[0.065, 1.000]
	16	[0.711, 0.852]	0.141	[0.130, 1.000]
	20	[0.769, 0.806]	0.038	[0.197, 1.000]

The $m = 8$ extremizers are plotted in Fig. S2. They are not fitted forecasts of best-of-100 truth; they are witnesses to nonidentification. The observed target lies inside every reported range, as it must because the empirical rank–truth law itself is feasible. Machine-readable endpoints, bootstrap critical values, grid checks, and all 1,000-bin extremizers are supplied in `results/coderm_partial_identification.csv`, `results/coderm_partial_id_grid_check.csv`, and `results/coderm_partial_id_worlds.csv`.

S10.3. Independent theorem checks

For $m = 1, \ldots, 8$ and four values of N per m, the reproducibility script performed three independent computations:

1. evaluated the closed form in Eq. S41;
2. constructed the barycentric interpolant at the U_m roots and integrated its absolute error by adaptive quadrature split at every root;
3. integrated $u^k\sigma_m(u)$ for $k = 0, \ldots, m-1$ to check audited-moment annihilation.

Across 32 cases, the largest absolute difference between items 1 and 2 was 1.887×10^{-15}. The largest absolute residual in item 3 was 3.331×10^{-16}.

Table S12: Representative exact-frontier checks. The complete 32-row table is `results/theorem_checks.csv`.

m	N	Closed form	Quadrature	Max moment residual
1	16	0.9999694824	0.9999694824	0
2	21	0.9952431821	0.9952431821	2.22×10^{-16}
4	31	0.9109153208	0.9109153208	4.16×10^{-17}
6	41	0.7510560990	0.7510560990	3.33×10^{-16}
8	51	0.5838695161	0.5838695161	1.11×10^{-16}

The numerical checks are not substitutes for the proof. They are designed to detect sign, indexing, interpolation-node, and phase-scaling errors in the implementation.

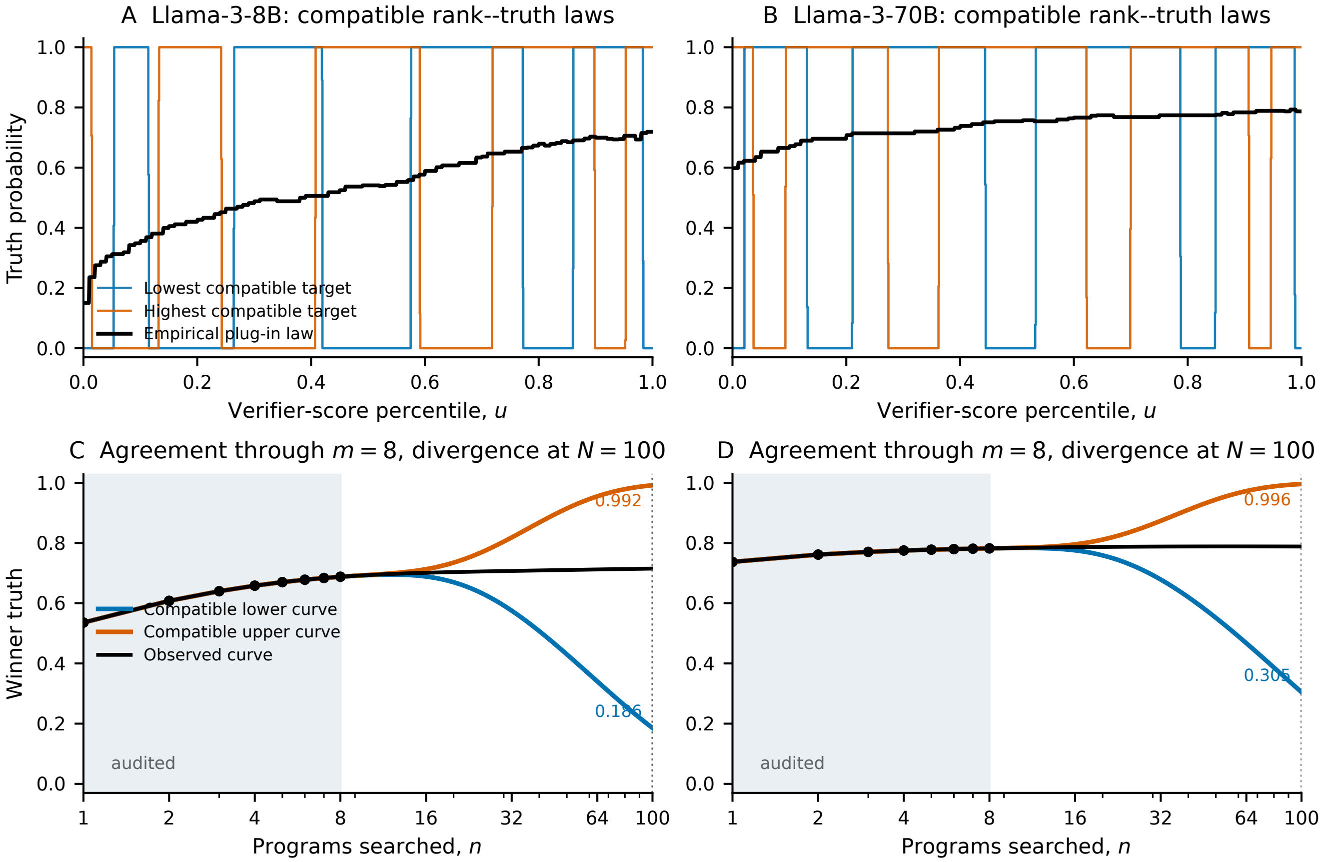

Figure S2: An empirical audited prefix does not certify the deployment target. (A,B) For each CodeRM model, linear programs construct bounded piecewise-constant rank–truth laws that match the empirical best-of-n means for every $n = 1, \ldots, 8$ while minimizing or maximizing best-of-100 truth; the black line is the empirical plug-in law. (C,D) The induced search curves coincide throughout the shaded audited range and then separate. At $N = 100$, the constructive plug-in ranges are $[0.186, 0.992]$ for 8B and $[0.305, 0.996]$ for 70B; observed values are 0.715 and 0.788. The grid laws are feasible continuum laws, so their separation is a witnessed lower bound on unrestricted empirical ambiguity.

S10.4. Reproduction workflow

The package contains:

- `code/reproduce_geometry_validation.py`: unrestricted and shape-restricted theorem checks, phase and finite-certificate planning tables, CodeRM reconstruction from derived numbers, QR-stabilized empirical partial-identification programs, finite masked-label acquisition, the frozen-rule discovery/held-out audit, Gaussian selection, intervention efficiency, and all deterministic figures;
- `code/solve_capped_tail_dual.py`: nested-bin primal–dual LPs, continuum scalar-threshold evaluation, optimizer coefficients, and percentile-scale L sensitivity;
- `code/analyze_search_spectrum.py`: pinned raw-data download, checksum verification, reference/deployment analysis, bootstrap, and processed tables;
- `code/analyze_split_robustness.py`: the 10-split robustness analysis;
- `code/analyze_coderm_search.py`: CodeRM checksum verification, streaming execution-matrix aggregation, exact search curves, task bootstrap, and derived candidate table;
- `data/processed/`: exact processed inputs, including 32,800 derived CodeRM score–truth rows, used by the lightweight reproduction;
- `results/`: machine-readable unrestricted and regularized theorem bounds, capped-tail primal–dual checks and optimizers, phase, finite certificate plans, Gaussian, CodeRM, partial-identification extremizers and grid checks, finite label-acquisition errors, discovery/held-out task assignments and errors, and design-efficiency outputs.

The lightweight script requires Python 3.12, NumPy, SciPy, and Matplotlib and does not download data or call a model API. The full GSM8K reconstruction additionally uses scikit-learn and downloads the two pinned raw files (approximately 18.5 GB combined). The CodeRM reconstruction reads the public 213-MB compressed output archive and the pinned annotation files. Every figure is deterministic. Bootstrap seeds are stated above; analytic theorem figures do not rely on Monte Carlo draws.

S11. Relation to adjacent literatures and contribution boundary

The mathematical ingredients have substantial precedents. The audit prefix is a truncated Hausdorff moment problem: classical work characterizes feasible moment sequences, while canonical-moment and Tchebycheff-system theory studies the geometry and extremal representations of the resulting moment spaces [8, 9]. Once $k_1, \ldots, k_m$ are recognized to span $\mathcal{P}_{m-1}$, optimal-recovery duality and canonical L^1 approximation give the unrestricted distance [4, 15]. Bounded-density moment theory supplies feasibility characterizations for the Lipschitz derivative measure [20]. The submitted objects are the diameters and residual support functionals induced by the stated best-of-N audit/deployment kernels after imposing bounded reliability, monotonicity, or a density cap. No general theorem about moment spaces, canonical representations, quotient duality, orthogonal polynomials, or moment representation is claimed as new.

Two close AI results mark the applied boundary. *ROC-n-reroll* gives verifiers with arbitrarily close finite best-of-N behavior and opposite infinite-compute limits [28], whereas the present unrestricted benchmark requires exact prefix equality and solves a specified finite target. *Evaluation Blind Spot* studies geometric recovery for static benchmark coverage [29]; the object here is selected-output truth under an explicit deployment kernel. These distinctions delimit the submitted object.

Closer statistical neighbors are Mallows's parent-distribution bounds from expected order statistics [30], Papadatos's expected-maxima and range sequences [13], and Okolewski and Papadatos's finite expected-order-statistic/truncated-moment connection [31]. They clarify feasibility and recovery. To our knowledge, they do not state the bounded-binary exact interval for a best-of-width reliability prefix, its low-width reliability-mean maximality, or the monotone and capped-Lipschitz residuals stated here. Huang et al. instead study coverage and optimal alignment under imperfect rewards [32]. Accordingly, the contribution claimed here is the exact validation object and its consequences, not a new characterization of expected order statistics.

S11.1. Theorem-by-theorem provenance

Table S13 gives the theorem-level boundary. Classical moment, approximation, and bounded-density results supply the mathematical ingredients [8, 9, 15, 20, 24]; finite expected-order-statistic work supplies the closest statistical representation results [13, 30, 31]; and recent best-of-N theory supplies the closest AI decision setting [32]. The third column states the remaining validation-specific contribution.

Table S13: Concise contribution boundary. "Submitted object" is validation-specific; the listed mathematical tools are established inputs.

Result	Classical input	Submitted object
Unrestricted benchmark	Hausdorff/canonical moment spaces [8, 9], quotient recovery [4], and canonical L^1 approximation [15]	Full bounded binary identified interval, attainable endpoints and interior laws, plus the finite theta phase law
Monotone frontier	Measure representation and uniform monomial approximation [24]	Exact reduction of compatible monotone validation worlds to $2E_m(u^N)$
Monotone Lipschitz frontier	Compact measure duality and bounded-density moments [20]	Exact capped-tail residual dual for the stated validation kernels
Endpoint bounds	Legendre orthogonality, Gauss recovery, and extreme-zero asymptotics	Explicit smooth compatible worlds and order-sharp fixed-L bounds
Scope and sampling	Span inclusion, concentration, and perturbation inequalities [27]	Width coverage separated from task precision; no new abstract recovery or concentration theorem is claimed

S11.2. Exact claim

The submitted claim is that the specified validation and deployment kernels produce the exact attainable identification objects in Table S13, including the monotone and capped-Lipschitz duals, the reliability-mean audit law, and the finite planning consequences. No general inverse-problem, moment, order-statistic, approximation, orthogonal-polynomial, or theta-function theorem is claimed. Complete proofs are given in Sections S1–S7, with executable numerical checks for every reported finite calculation.

The empirical role is narrower. Independent units are 127 problems and 164 tasks; candidate outputs repeat within them. The analyses show target change, reversals, compatible witnesses, and a within-pool acquisition contrast, not prevalence, extremal-world frequency, or prospective effectiveness. The 82/82 split protects held-out labels from rule selection but remains a public-data replay.